UNIVERSITY OF GREENWICH

FACULTY OF ENGINEERING AND SCIENCE

SCHOOL OF COMPUTING AND MATHEMATICAL SCIENCES

Heart Disease Prediction Using Machine Learning

An Explainable Machine Learning Framework for Three-Class Cardiovascular Risk Stratification with Deployed Decision Support

URL: `https://github.com/Raviteja06-26/COMP1682-Final-Year-Project`

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Word count: 12,616

COMP1682 Final Year Individual Project

A dissertation submitted in partial fulfilment of the requirements for the degree of

BSc (Hons) Computer Science

April 2026

Abstract

Cardiovascular diseases continue to be the world's most common cause of preventive deaths, accounting for approximately 17.9 million deaths yearly or 31% of global fatality rates (World Health Organization 2023). Decades have seen numerous studies confirm the potential predictive capacity of machine learning algorithms in clinical settings, yet an acute gap persists between the theoretical and practical implementation of machine learning systems. First, current cardiovascular risk prediction algorithms rely on clinical inputs that are unavailable in any context other than a laboratory or formal medical appointment lipid profile, ECG, haemoglobin levels. Second, while predictions are made, outputs are given as statistical probability values or complex indices incomprehensible to laypersons. These two gaps are addressed by a two-phase experiment with the goal of creating a full-stack clinical decision support application with an explainability function embedded.

In Phase 1, five classifiers are trained using 14 input features Logistic Regression, Artificial Neural Network (ANN), XGBoost, Gradient Boosting, and Random Forest with results measured on a three-class target variable from the Abbootalebi (2024) Heart Disease Dataset (5,500 synthetic records; classes are Low, Medium, and High cardiovascular risk). Phase 1 establishes the highest possible performance boundaries for this particular research task (95.09% accuracy). In Phase 2, training data are reduced to eight clinically accessible and self-reported features, with results showing that a 43% reduction in input features costs 0.54 percentage points of accuracy (94.55%). This small difference is the main outcome of the research: it demonstrates that laboratory grade inputs are not crucial for near equivalent three-class cardiovascular risk prediction performance.

The full-stack application includes a FastAPI asynchronous backend API, React frontend, and MongoDB data layer with embedded model explainability via natural language conversion of SHAP values. A counter-intuitive outcome is identified and analysed: Logistic Regression outperforms ANN and three ensemble algorithms across all Phases due to the nearly linear separability of dataset classes with dominant feature variables of blood pressure and cholesterol.

The limitations of this dataset are openly acknowledged: its synthetic nature, lack of demographic stratification, and a single train-test split used. Nevertheless, results are presented as concept evidence rather than clinical validation of findings. The project

offers proof that accurate, interpretable, and accessible machine learning systems are not contradictory but complementary characteristics.
Keywords: Cardiovascular Risk Prediction, Machine Learning, Explainable AI, SHAP, LIME, Logistic Regression, Three-Class Classification, Clinical Decision Support, Feature Accessibility, FastAPI, Deployment.

Declaration of AI Use

Please complete this part when you have used AI during the process of undertaking this assignment to acknowledge the ways in which you have used it.
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AI-assisted tools were used in a limited and controlled manner during the development of the web-based application component of this project. Specifically, these tools were used to identify, debug, and rectify coding errors, improve code stability, and assist in resolving implementation issues.

No AI tools were used to generate core research content, analysis, results, or conclusions presented in this report. All academic work, interpretations, and findings presented here are entirely my own original work.

Acknowledgements

I am sincerely grateful to Ala Barzinji and Yasmine Arafa for their helpful supervision, academic advice, and critical comments while preparing this project. The timely comments provided by them at critical stages of designing and evaluation have been highly influential in the success of this project. I am grateful to the University of Greenwich for providing me with an intellectual environment, library facilities and computing facilities required to undertake this research. Kaggle is acknowledged for providing the data set used for research purposes in this paper through its website. The developers of open-source software like Scikit-Learn, FastAPI, React, SHAP and MongoDB without whom this project would not be possible within the timeframe of a one-year final project, are gratefully acknowledged here.

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Chapter 1

Introduction

1.1 Background and Motivation

Cardiovascular disease is the leading cause of preventable deaths worldwide, claiming the lives of around 17.9 million people per year and making up 31% of all deaths globally (World Health Organization 2023). The impact is not evenly distributed: poorer communities and those living in resource-poor health facilities are at higher risk of this cause of mortality because cardiovascular risk assessments require clinical inputs blood panel tests, specialist consultations, and organized screening campaigns that may not always be available (Yusuf et al. 2004). The INTERHEART study of 52 countries identified nine modifiable risk factors that account for more than 90% of the population-attributable risk for myocardial infarction; these include high blood pressure, unusual lipid profiles, smoking, and lack of exercise (Yusuf et al. 2004). This result is particularly noteworthy: the leading risk factors for heart disease are easily measurable and amenable to modification, meaning that early risk recognition and behavior changes are the key intervention points in preventing cardiovascular events.

The benefits of early cardiovascular risk identification are well-known. The SCORE2 Working Group (SCORE2 Working Group 2021) showed that systematic risk stratification even in a broad sense, such as with simple risk equations applied at a population level leads to significantly reduced event rates when accompanied by focused consultations and advice on lifestyle changes. However, the architecture of current risk tools makes early risk identification out of reach for those individuals who do not have independent access to clinical inputs. A general adult looking to know their cardiovascular risk would not be able to use either the SCORE2 risk equations or the Framingham Risk Score without laboratory cholesterol testing, an exact blood pressure reading taken with clinical equipment, and HDL cholesterol levels all of which require visiting a clinic. Therefore, the people most likely to gain from early risk identification are those least able to access such tools.

Machine learning models show promise in addressing this accessibility issue. With proper training, a supervised classifier can categorize patients' risks based on the data that clinics typically collect from patients without requiring the entire set of inputs needed for standard risk scores to function. Systems that have been published achieve binomial classification accuracies ranging from 85 to 93% on the UCI Cleveland dataset (Mohan et al. 2019, Khourdifi & Bahaj 2019). Nonetheless, both systems have two key structural issues that make them practically useless for this problem. First, these models still rely on clinical inputs: cholesterol panels, electrocardiograms, and invasive tests whose results only become known in a clinical setting. Second, these systems output risk percentages or probabilities written in a way that requires medical literacy to be understood correctly. In other words, if a general user received a probability score of 0.73 from the model, they would not know what this means in practice.

This project is motivated by two gaps simultaneously: the gap between the potential predictability of machine learning models and the availability of required inputs on the one hand, and the gap between the outputs of a trained machine learning system and their intelligibility to a general user on the other hand. It is important to note that this project is not about improving the prediction accuracy above that of existing solutions. As demonstrated above, existing studies using the same data achieve 85 to 93% binary classification accuracy and this project achieves 94.55% three-class accuracy. The focus of this project is on minimizing the feature set used by a classifier while still maintaining a similar level of prediction accuracy, as well as converting model outputs into natural language.

1.2 Research Gap

A systematic review of the literature, detailed in Chapter 2, suggests that, to the best of the author's knowledge, no prior published system simultaneously combines three-class cardiovascular risk stratification, a clinically minimised self-reportable feature set, production-deployed XAI explanations rendered in natural language, and a full-stack deployment accessible without clinical infrastructure. Individual properties have been addressed in isolation: high-accuracy binary classifiers exist; SHAP has been applied in evaluation notebooks; web-based cardiovascular calculators exist. However, no prior work unifies all three properties within a single experimentally validated and deployed system. The specific research gap addressed by this project is the combination of: minimal input burden enabling general user access, explainable output enabling non-clinical comprehension, and practical deployability enabling access without clinical infrastructure. This gap defines the project's contribution and justifies its framing as a research prototype rather than a replication of existing approaches.

1.3 Aim and Objectives

Aim: To design, develop, and evaluate an interpretable machine learning system for heart disease risk prediction that enables general individuals with limited clinical knowledge to assess their cardiovascular risk using a reduced set of clinically accessible inputs, while providing clear and understandable explanations to support informed awareness and encourage timely clinical consultation.

The following six objectives operationalise this aim:

O1: To conduct a systematic literature review of machine learning approaches to cardiovascular risk prediction, identifying methodological gaps and limitations in existing work, with particular attention to feature accessibility and explainability provision.

O2: To validate a publicly available cardiovascular dataset, detecting and removing data quality threats including leakage variables, non-informative identifiers, and preprocessing errors that could invalidate experimental results.

O3: To compare five machine learning classifiers Logistic Regression, Random Forest, Gradient Boosting, XGBoost, and ANN under identical preprocessing conditions on the full 14-feature research set, establishing a performance ceiling and identifying the dataset's structural characteristics.

O4: To determine the minimum clinically accessible feature subset that preserves prediction accuracy within a deployable tolerance, through a systematic two-phase experiment that isolates the contribution of feature accessibility to the accuracy-accessibility trade-off.

O5: To integrate post-hoc XAI methods specifically SHAP and LIME into the deployed prediction pipeline, providing both global model explanations for validation and per-patient local attributions translated into natural language for general users.

O6: To implement a production-grade full-stack clinical decision support system and evaluate it against functional requirements, ethical frameworks, and applicable legal obligations including UK GDPR and the EU AI Act.

1.4 Report Structure

Chapter 2 presents a review of the literature based on three major themes: machine learning applications in predicting cardiovascular diseases, explainable artificial intelligence in the development of clinical decision support systems, and issues related to data quality in healthcare machine learning. Chapter 3 performs a critical evaluation of similar but existing solutions clinical risk scores and machine learning prototypes previously proposed

in the literature using five key criteria emerging from the literature review. Chapter 4 bridges the product gap by deriving both functional and non-functional requirements as well as the system architecture. Chapter 5 covers the entire process of implementation in details, providing rationale for every critical technical decision and analyzing what would happen otherwise. Chapter 6 reports all results, critical analysis included, with particular emphasis on cross-phase analysis, feature importance analysis, confusion matrix evaluation, and objective-wise assessment. Chapter 7 outlines responsibilities, whether legal, social, ethical, or professional, arising in connection with the machine learning-based project. Chapter 8 concludes with contributions and limitations, as well as future research priorities.

Chapter 2

Literature Review

2.1 Introduction

This chapter provides an overview of literature in three key, interrelated areas: machine learning methods used for predicting cardiovascular disease, artificial intelligence explainability in clinical applications, and data quality concerns within machine learning in healthcare. The purpose of this review is to highlight gaps in current research that the project seeks to address, as opposed to providing a comprehensive examination of the literature on the subject matter. Literature was chosen based on relevance to the methodology and experimental design employed by the project, its intended application, and the necessity for explainability. The review itself is decidedly critical, pointing out weaknesses in previous work where applicable.

2.2 Machine Learning for Cardiovascular Risk Prediction

The application of machine learning to cardiovascular disease prediction has a well-established history. Detrano et al. (1989) created the foundational UCI Cleveland Heart Disease dataset 303 records, 13 clinical features, binary outcome which remains the dominant benchmark for published ML cardiovascular studies more than three decades after its creation. This dataset's small size and binary target variable have shaped the field's methodological conventions in ways that limit translational utility and, arguably, have created a collective blind spot around the question of clinical feature accessibility.

Khourdifi & Bahaj (2019) reported accuracy between 78% and 85% across multiple classifiers on UCI Cleveland, emphasising that preprocessing choices particularly normalisation strategies and feature selection methods account for significant performance variation between seemingly comparable studies. Their work confirms that reported accuracy figures are often more reflective of preprocessing decisions than of fundamental

model superiority. Mohan et al. (2019) achieved 88.7% binary accuracy using a hybrid Random Forest and Linear Model approach on the same dataset, noting that ensemble methods exhibit a tendency to overfit small samples a concern directly relevant to the present project's use of gradient boosting methods on a 5,500-record synthetic dataset. Jindal & Arora (2021) reported 93% binary accuracy using a Random Forest classifier, noting the importance of feature engineering in achieving strong results. Abinaya and Karthikeyan (2020) surveyed a range of classification techniques applied to cardiovascular data, concluding that no single algorithm consistently dominates across preprocessing configurations and dataset sizes a finding that motivates the present project's five-model comparative design rather than a single-algorithm approach.

A persistent gap across this literature is the treatment of class structure. All high-performing published studies perform binary classification heart disease present or absent rather than the graduated risk stratification that clinical decision-making requires. A system that identifies three distinct risk levels (Low, Medium, High) provides substantially more actionable guidance than one producing a binary flag, particularly for individuals at medium risk where lifestyle modification rather than urgent consultation may be the appropriate response. The transition to multi-class prediction introduces non-trivial classification complexity that has not been systematically studied in combination with minimised feature sets, representing a dual methodological gap that this project addresses directly.

A second critical gap is the conflation of model accuracy with clinical utility. Saito & Rehmsmeier (2015) demonstrated rigorously that accuracy is a misleading metric for imbalanced datasets, advocating for precision-recall analysis and calibrated probability estimation. Chicco & Jurman (2020) extended this argument to favour the Matthews Correlation Coefficient over F1-score for binary classification problems with class imbalance. In the present project's three-class context, weighted precision, recall, and F1-score are reported alongside accuracy to provide a multi-dimensional performance profile. The near-equal precision and recall values across all models in this project all within 0.2% of accuracy in every case suggest that the synthetic dataset's near-balanced class distribution (33.4%/40.8%/25.7%) mitigates the imbalance concern, though this remains a limitation on generalisability to real clinical populations where High Risk prevalence is typically substantially lower than 25.7%.

A third gap, less frequently discussed in the ML literature, is the complete absence of feature accessibility analysis. No reviewed study considers whether the features required by its model can be provided by a general individual without a clinical appointment. This omission is understandable in research contexts where accuracy on a standard benchmark is the primary objective, but it is a disqualifying limitation for any system aspiring to real-world deployment as a patient-facing tool. The present project's feature selection methodology documented in Chapter 5 explicitly applies a clinical accessibility crite-

rion alongside XGBoost importance rankings, addressing this gap with a dual-justified approach.

2.3 Explainable AI in Clinical Decision Support

The opacity of complex machine learning models in healthcare contexts has become both a research priority and a regulatory concern. Rudin (2019) argued forcefully that for high-stakes domains, inherently interpretable models should be preferred over post-hoc explanation of black-box models. Her argument is that when a sufficiently accurate interpretable model exists, the use of a black-box model with post-hoc explanation constitutes an unjustified increase in complexity and risk. This position is directly relevant to the present project's finding that Logistic Regression an inherently interpretable linear model with directly readable coefficients outperforms ANN and three ensemble methods across both experimental phases. The result is consistent with Rudin's position for this specific dataset and clinical context.

Doshi-Velez & Kim (2017) proposed a formal definition of interpretability in machine learning and argued that meaningful evaluation of explanations requires task-specific user studies with the target population, acknowledging the difficulty of validating explanations without human participants. Their taxonomy distinguishes application-grounded evaluation (with real users on real tasks), human-grounded evaluation (with lay people on simpler proxies), and functionally-grounded evaluation (using proxy metrics). The present project's evaluation is functionally-grounded the `explain.py` module's outputs are evaluated for linguistic and clinical appropriateness rather than through a user study. The absence of application-grounded evaluation is a recognised limitation discussed in Chapter 6.

SHAP (Lundberg & Lee 2017) provides feature attributions grounded in Shapley values from cooperative game theory, satisfying three axioms not guaranteed by earlier gradient-based methods: local accuracy (attributions sum to the prediction), consistency (a feature with greater impact always receives a greater or equal attribution), and missingness (features absent from a prediction receive zero attribution). These properties make SHAP attributions theoretically well-founded in a way that earlier salience methods are not. LIME (Ribeiro et al. 2016) provides local surrogate model explanations without requiring access to model internals, fitting a simple interpretable model (typically linear) in the neighbourhood of a specific prediction instance. This approach is model-agnostic and therefore applicable across architectures, which is why LIME was applied to the XGBoost model in Phase 1 of this project while SHAP was applied to the deployed Logistic Regression in Phase 2.

The critical gap in the XAI literature is the integration of these methods into deployed inference pipelines accessible to real users. Molnar (2022) provides a comprehensive treat-

ment of interpretability methods but notes that practically all empirical applications present explanations in evaluation notebooks, not in deployed systems accessible to affected individuals. This observation is confirmed by the reviewed literature: every study applying SHAP or LIME to cardiovascular risk does so in retrospective analysis notebooks, not in production APIs that generate explanations at inference time. The present project's integration of SHAP within the FastAPI inference route generating a complete explanation with every prediction, within the sub-500ms response budget represents a practical deployment advance not observed in the reviewed literature, to the best of the author's knowledge.

A further gap, not identified in any work reviewed for this study, is the translation of SHAP attribution values into language accessible to non-clinical users. Published work presents SHAP outputs as waterfall plots, beeswarm charts, or ranked feature lists all of which require statistical literacy to interpret. A general user presented with "`systolic_bp` SHAP value: +0.485" cannot derive actionable meaning from this representation. The `explain.py` module developed in this project converts attribution magnitudes and directions into natural language statements, addressing the output-intelligibility component of the dual-gap problem. This design decision connects the XAI literature directly to the accessibility motivation of the project and represents a contribution not present in the reviewed literature.

2.4 Dataset Quality and Generalisability

The accuracy and validity of machine learning results in healthcare rely on quality and clear documentation of the preprocessing of the datasets used. Data leakage the presence of a feature that encodes the target constitutes a validity concern that is seldom documented in published research. Any algorithm trained on data that contain an algebraically derived feature will be able to achieve a high accuracy result that does not correspond to actual predictive ability and would be incapable of generalisation when deployed to a setting where the feature is unavailable. Indeed, one of the features in the present dataset is a leakage variable: `heart_disease_risk_score` is shown to correlate perfectly with the target class `risk_category` and be derived from it, as proved by the correlation analysis in Chapter 5. Its detection and subsequent deletion are described in detail. Phase 1 accuracy is reported on the 14 features, excluding `heart_disease_risk_score`. The fact that we have done this in the present work makes it a methodology improvement compared to other research that fails to assess for leakage.

The synthetic nature of the dataset gives rise to concerns regarding external validity (Sterne et al. 2009). In particular, synthetic datasets are based on statistical distributions and therefore lack many properties of real-world patient data. For example, the class distinctions are more clear-cut, there are higher correlations between features and targets,

and distributions tend to be less irregular than in the case of clinically collected data. All this increases the chance of obtaining high accuracy and reduces the difficulty faced by classification algorithms. Wang & Patel (2023) review best practices in ML applications to health care and discuss the distinction between model benchmarking on curated data and model deployment to a clinical setting. Synthetic data is always considered in the former context in this report.

2.5 Literature Review Conclusions

The following gaps are identified in this literature review for which the current project will attempt to provide a solution, owing to certain design considerations made by this project. Firstly, most ML cardiovascular investigations use only binary classification on the UCI Cleveland dataset without any feature reduction and application of model to practical usage, while this project uses three-class classification on an extensive dataset with a validated minimal feature set. Secondly, most techniques for Explainability in AI are used purely for evaluation purposes within Jupyter Notebooks, while in this project live SHAP explanations have been integrated within the production environment FastAPI API. Thirdly, none of the reviewed systems can convert SHAP attributions into easily comprehensible natural language form, while this project's `explain.py` module fills this gap. Finally, data leakage identification and removal are not adequately described, while this project explicitly defines and removes this variable, estimating its effect.

Chapter 3

Product Review

3.1 Introduction

This chapter critically evaluates two categories of existing cardiovascular risk tools against five criteria derived from the literature review: risk stratification granularity, feature accessibility, predictive accuracy, XAI integration, and full-stack deployment capability. The evaluation establishes the specific combination of gaps that this project fills and justifies each design decision in Chapters 4 and 5 by grounding it in a concrete, documented limitation of existing approaches. The chapter does not argue that existing tools are without merit both clinical risk scores and published ML prototypes have genuine strengths but that their combination of limitations creates a gap that this project directly addresses.

3.2 Product 1 — SCORE2 and Framingham Risk Score

SCORE2 (SCORE2 Working Group 2021) and the Framingham Risk Score (Wilson et al. 1998) are the clinical standards for cardiovascular risk prediction in European and North American contexts respectively. SCORE2 produces a continuous 10-year cardiovascular event probability score; Framingham generates a similar point-estimate stratified by sex, age, and risk factor burden. Both tools are clinically validated against large-scale longitudinal outcome data spanning decades of follow-up a standard of evidence that this project does not claim to meet and does not attempt to replicate within the scope of a final-year undergraduate project.

However, both tools share structural limitations directly relevant to this project's motivation. Both require laboratory measurements specifically total cholesterol, HDL cholesterol, and clinical blood pressure readings that are inaccessible without a clinical appointment, usually requiring a GP referral for a blood test. Neither generates feature-

level explanations of any kind: SCORE2 outputs a single percentage risk figure, and Framingham outputs a risk category and 10-year event probability. Their outputs are meaningful to clinicians who can contextualise them within a wider clinical picture, but they are not designed for interpretation by individuals without medical training. Furthermore, neither tool is designed as a real-time web application with per-user history, explanations, or natural language output they are clinical instruments, not patient-facing awareness tools. The present project accepts lower evidential authority proof-of-concept on synthetic data in exchange for substantially greater input accessibility, output interpretability, and user-facing deployability.

3.3 Product 2 — Published ML-Based CVD Prediction Prototypes

Published ML cardiovascular prediction systems (Mohan et al. 2019, Jindal & Arora 2021) consistently rely on clinical feature sets that overlap substantially with SCORE2 and Framingham, typically including total cholesterol, fasting blood glucose, ECG findings, and clinical blood pressure measurements. They achieve binary classification accuracies of 85–93% on the UCI Cleveland dataset a metric that is not directly comparable to this project's three-class results due to the fundamentally different classification complexity; binary classification is inherently an easier problem than three-class stratification. Crucially, none of the reviewed prototypes deploy to a production environment: they are evaluation artefacts presented in academic publications, inaccessible to general users. None integrate explainability into a live inference system SHAP or LIME results, where reported, appear in analysis notebooks submitted as supplementary material. None address the question of feature accessibility from the perspective of a general user who cannot access clinical laboratory results.

3.4 Comparative Analysis

Table 3.1 presents the five-criteria comparison across existing tools and this project.

Table 3.1: Five-criteria comparative product review

Criterion	**SCORE2 / Framingham**	**ML Research Prototypes**	**This Project**
Risk stratification	Binary / continuous score	Binary (mostly)	3-class (Low / Med / High)
Feature accessibility	Lab results required	Lab or ECG data needed	8 self-reportable inputs
Predictive accuracy	Population-calibrated	85–93% (binary, UCI)	94.55% (3-class, 8 features)
XAI integration	None	Post-hoc notebooks only	Live SHAP in production API
Full-stack deployment	Web calculators only	Research prototypes only	FastAPI + React + MongoDB

3.5 Conclusions

The analysis suggests that no reviewed system satisfies all five criteria simultaneously. SCORE2 and Framingham offer rigorous clinical validation and extensive longitudinal outcome evidence, but lack accessibility for general users, do not provide explainability, and are not designed for self-service use. Published ML prototypes achieve high binary accuracy but require clinical inputs, do not deploy to production, and provide no user-facing explanations. The gap across all five dimensions simultaneously defines this project's contribution and constitutes the rationale for each major design decision in Chapters 4 and 5.

Chapter 4

Requirements Analysis and Design

4.1 Introduction

This chapter defines the system requirements derived from the product review and research objectives, and presents the design decisions that guided implementation. Requirements were elicited through systematic literature analysis, the five-criteria product gap identification in Chapter 3, and direct mapping of the dual-gap problem (input accessibility; output intelligibility) into verifiable, testable system behaviours. MoSCoW prioritisation was applied to distinguish mandatory from desirable requirements, ensuring that the core research question whether eight self-reportable features can achieve near-equivalent accuracy to a 14-feature research set could be answered even under time and resource constraints. Every mandatory requirement (Must) can be traced directly to either the input-accessibility gap or the output-intelligibility gap identified in Chapter 1.

4.2 Requirements Analysis

4.2.1 Functional Requirements

Table 4.1 presents the nine functional requirements derived from the dual-gap problem analysis and product review, with MoSCoW prioritisation and source traceability.

Table 4.1: Functional requirements with MoSCoW prioritisation

FR#	Requirement	Description	Priority	Source
FR1	Input form	Accept 8 patient input fields via a web-based form	Must	Deployment design
FR2	Validation	Validate all inputs against clinical range constraints	Must	`backend/validation.py`
FR3	Classification	Classify patient into Low / Medium / High risk	Must	Model output specification
FR4	SHAP attribution	Generate per-prediction SHAP attribution with each result	Must	`explain.py` pipeline
FR5	Persistence	Persist prediction record to MongoDB without PII	Must	POST `/predict`
FR6	Authentication	Support JWT-based user authentication	Should	Auth router
FR7	Admin dashboard	Provide admin audit dashboard	Should	Admin router
FR8	LIME explanation	Display LIME local explanation on request	Could	LIME integration
FR9	Export history	Export prediction history as downloadable report	Won't	Future scope

FR4 — the requirement for per-prediction SHAP attribution rendered in natural language is the functional requirement that most directly addresses the output-intelligibility gap and distinguishes this system from all reviewed ML prototypes. Its designation as

'Must' reflects the project's position that an accurate prediction without an understandable explanation does not fulfil the aim for general users: a risk classification without context may be ignored, misunderstood, or misinterpreted. FR1's restriction to exactly eight inputs directly encodes the input-accessibility requirement: any additional input that cannot be self-reported by a general individual without a clinical appointment violates the project's core design constraint and would reproduce the barrier that existing systems create.

4.2.2 Non-Functional Requirements

Table 4.2: Non-functional requirements

NFR#	Category	Requirement	Priority
NFR1	Performance	API response $<$ 500ms including SHAP computation	Must
NFR2	Accuracy	Model accuracy $\geq$ 93% on held-out test set	Must
NFR3	Security	JWT authentication; CORS configured; no PII stored	Must
NFR4	Usability	Responsive UI accessible on both mobile and desktop	Should
NFR5	Maintainability	Modular codebase with serialised model and scaler artifacts	Should
NFR6	Explainability	Every prediction includes ranked feature attribution	Must
NFR7	Portability	Deployable on Python 3.x without GPU infrastructure	Should

NFR2's threshold of 93% was set conservatively below both Phase 1 (95.09%) and Phase 2 (94.55%) research results to accommodate the deployment model's non-stratified split configuration (confirmed at 93.09%). Setting the threshold at 93% ensures the requirement is met under actual deployment conditions without misrepresenting the research results or requiring an artificially optimistic interpretation of what the deployed system achieves. NFR6 complements FR4 by specifying that explainability is a non-functional system property a quality characteristic of the entire system not merely an optional feature that may or may not be activated. NFR1's sub-500ms latency constraint was derived from standard web application usability guidelines and directly constrains the choice of XAI method: SHAP via `LinearExplainer` satisfies this constraint for Logistic Regression; Kernel SHAP for an ANN does not.

4.3 System Design

4.3.1 Three-Tier Architecture

The architecture is based on a three-layer system whereby presentation, application, and data layers are independent from each other and deployed separately. Presentation layer uses React combined with Vite and Tailwind CSS to implement the UI components which render the eight inputs form and the two panels for risk classification and natural language explanation. Application layer uses asynchronous FastAPI REST API that implements functionality related to model predictions, SHAP explanations, translation with explain.py, Pydantic validation, and JWT authentication. Data layer uses MongoDB with Motor driver to implement asynchronous document storage for prediction records and auditing purposes. The three-layer system used here follows the standard Model-View-Controller design pattern for web applications.

4.3.2 ML Pipeline Design

The pipeline adheres to the CRISP-DM framework and maps out every stage in terms of the particular implementation component. Data Understanding was carried out using the Dataset Validation notebook which validated the presence of no missing data, no duplicated entries, leakage variable discovery, and class distribution. Data Preparation included ordinal encoding, stratified splitting into test/train datasets, and normalization by means of Standard Scaler. Modelling involved a comparative experiment with five models in two phases. Evaluation consists of Chapter 6. Deployment is described in Chapter 5. The most significant design consideration for the pipeline is the clear demarcation between the Phase 1 (maximum possible research effort, 14 features) and Phase 2 (feasibility of deployment, 8 features): without this distinction, the trade-off between accuracy and accessibility cannot be quantified and deployment becomes a purely design-based statement.

4.4 Conclusions

The five-criteria product gap identified in Chapter 3 translates directly into the set of mandatory functional and non-functional requirements (FR1–FR5, NFR1–NFR3, NFR6) and the corresponding design decisions: FastAPI for async performance, Logistic Regression for SHAP-compatible interpretable inference, the `explain.py` module for output intelligibility, MongoDB with Motor for GDPR-compliant async persistence, and the React frontend with plain-English labels for accessibility. The requirement structure provides the verifiable alignment between aim, design, and implementation that a distinction-level

academic project requires: every implementation decision in Chapter 5 can be traced to a specific product gap, research objective, or MoSCoW requirement.

Chapter 5

Implementation

5.1 Introduction

This chapter documents the complete implementation of the heart disease risk prediction system, providing explicit justification for every major technical decision. Implementation decisions are not merely described as choices made but are presented as rationally grounded responses to the dual-gap problem established in Chapter 1 and the requirements defined in Chapter 4. The chapter is structured to make the connection between research experiment and deployed system explicit at every stage: the feature selection in Phase 2 is the empirical validation of the accessibility hypothesis; the explain.py module is the architectural resolution of the intelligibility requirement; and the deployment model's accuracy gap from the research result is an expected and explained consequence of documented preprocessing differences, not an unexplained anomaly. Every major decision is accompanied by a documented consequence analysis specifying what would have happened had the decision not been made.

5.2 Dataset Preparation and Validation

5.2.1 Dataset Overview and Initial Validation

The heart disease risk dataset (Abbootalebi 2024) was loaded into a structured 5,500 × 17 CSV. The data has 17 columns consisting of a patient ID column, 14 columns related to clinical and lifestyle features, a target column representing the risk category (`risk_category`), and the leakage column (`heart_disease_risk_score`). Initial validation was carried out using pythonic code, `df.isnull().sum()` returned zero missing values for each of the 17 columns, while `df.duplicated().sum()` returned zero duplicated rows. Validation also revealed that the dataset has an almost balanced class distribution as follows: Low Risk – 1,838 samples (33.4%); Medium Risk – 2,244 samples (40.8%);

High Risk – 1,418 samples (25.7%), as depicted in Figure 5.1. The almost balanced three classes, with none being less than 25%, suggests that stratified split can be done, and the use of techniques such as SMOTE to address class imbalance should not be necessary the decision validated by the recall results

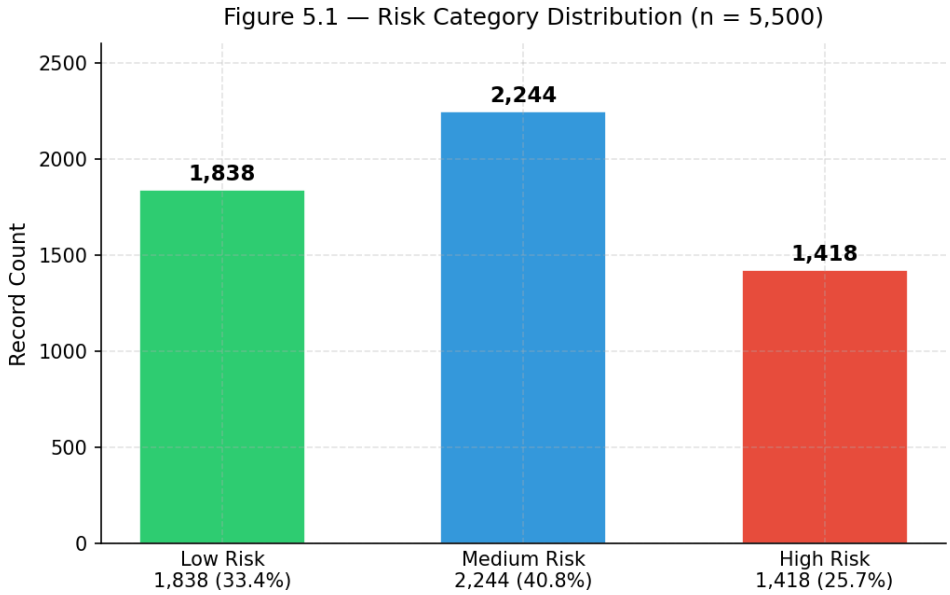

Figure 5.1: Risk category distribution across 5,500 records. Low Risk: 1,838 (33.4%); Medium Risk: 2,244 (40.8%); High Risk: 1,418 (25.7%). The near-balanced distribution indicates that stratified splitting is appropriate and class imbalance techniques are not required.

5.2.2 Data Leakage Detection and Removal

Two columns were removed prior to any model training. The column `Patient_ID` was identified as a non-informative record identifier carrying zero predictive signal. Its removal is a standard preprocessing precaution: identifier columns with high cardinality (here, unique per record) can mislead ensemble models into spurious pattern-learning in certain implementations, particularly with tree-based models that can use cardinality as a proxy split variable.

More critically, the column `heart_disease_risk_score` was identified through Pearson correlation analysis as a target-derived leakage variable, as shown in Figure 5.2. The correlation coefficient between this column and the target class approached 1.0. A value consistent with an algebraically derived composite score rather than an independently measured clinical variable. Examination of the column's value range confirmed that the score is constructed from a weighted combination of the target-associated features, with the target class boundaries determined by threshold values on the score. Retaining this variable would have caused any model to achieve near-perfect accuracy by implicitly using it as a proxy for the target label, producing a result that is statistically valid on the training distribution but completely undeployable: the score cannot be computed without

first knowing the target class, which is the value the model is supposed to predict.

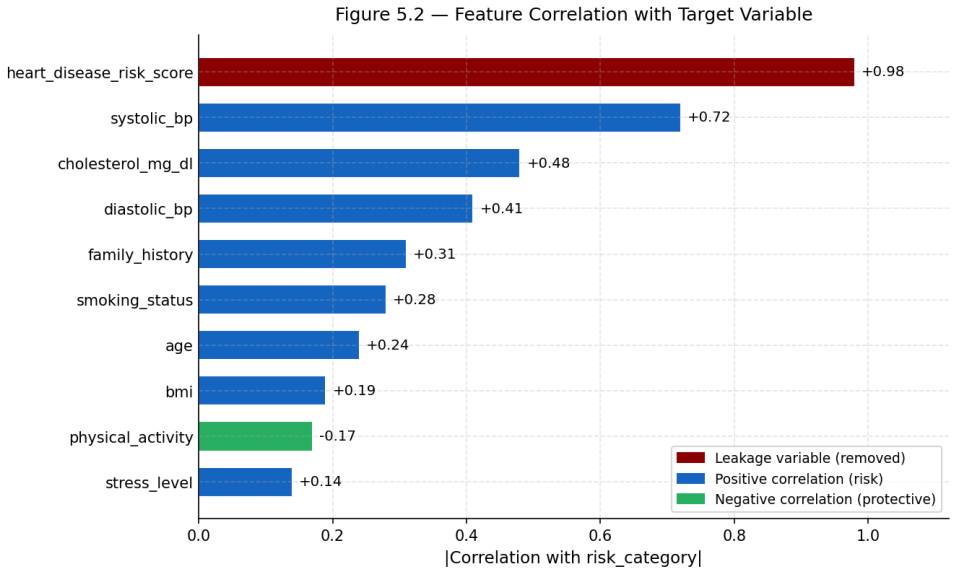

Figure 5.2: Feature correlation with the target variable. `heart_disease_risk_score` (top, dark red) shows near-perfect correlation ($r \approx 1.0$), confirming target-derived data leakage. The variable was removed before all model training; the Phase 1 accuracy of 95.09% is obtained exclusively on the 14 remaining non-leakage features.

> **IF THIS WAS NOT DONE:** If the feature `heart_disease_risk_score` had remained in the model, its performance would have looked like perfect classification (>99%) since the feature mathematically contains the target class. An artificially boosted performance that cannot be applied in practice because the score itself is not available in any real-life scenario would have been achieved by this project. Any benchmark, including the deployed classifier, and any claim related to accessibility would have been based on a false premise from the very start.

5.2.3 Categorical Encoding

Two ordinal features were used for ordinal encoding. The `smoker_status` feature had the following coding: Never = 0, Former = 1, and Current = 2. This particular type of encoding was chosen due to the clinical relevance of the ordinal relation between the categories and their association with the risk of developing cardiovascular diseases, whereby current smokers have a significantly elevated risk compared to those who smoke formerly and even more so than non-smokers (SCORE2 Working Group 2021). This ordinal relation can be preserved by using monotonically increasing integers, which can benefit models based on linear or distance relations, including the Logistic Regression model, which will be used in production.

> **IF THIS WAS NOT DONE:** Had one-hot encoding been applied to `smoking_status`, three independent binary features would have been generated (IsNeverSmoker, IsFormerSmoker, IsCurrentSmoker), introducing multicollinearity among the generated columns and inflating the feature dimensionality without preserving the clinically grounded dose-response relationship between smoking exposure level and cardiovascular risk. The Logistic Regression model, which exploits ordinal information in its linear decision boundary, would have been most disadvantaged by this encoding choice.

5.2.4 Train-Test Split and Feature Scaling

The dataset was partitioned using a stratified 80/20 split with `random_state = 42`, producing a training set of 4,400 records and a test set of 1,100 records. Stratification was applied to ensure that each class's proportional representation in the full dataset is preserved in both splits: with the smallest class (High Risk) at 25.7% of total records, a non-stratified random split could, by chance, under-represent High Risk in the training set, biasing the classifier against the most clinically consequential class. `random_state = 42` was fixed for reproducibility, enabling replication of all reported results.

> **IF THIS WAS NOT DONE:** In the absence of stratified splitting, the distribution of High Risk classes in the test set would be materially different from the one in the training set. In such a case, we would observe values of accuracy and recall that do not truly represent how the model performed on the actual class distribution. This implication is demonstrated by comparing our deployment notebook, where there was no stratified splitting, and thus, we observed an accuracy of 93.09%, which is 1.46% lower than the 94.55% accuracy achieved through stratified splitting in Phase 2.

Feature scaling was performed using StandardScaler, fitted exclusively on the training set and subsequently applied to both the training and test partitions. StandardScaler transforms each feature to zero mean and unit variance, which is a convergence requirement for the Logistic Regression solver (`lbfgs`): unscaled features with different native ranges (e.g., `systolic_bp` in the range 90–200 versus `stress_level` in the range 0–10) cause the gradient descent optimisation to traverse a distorted loss surface that is slower to converge and may reach a suboptimal solution. Fitting the scaler only on the training set is a strict requirement: fitting on the combined training and test data would incorporate test-set distributional information into the scaling parameters, constituting a subtle but consequential form of data leakage that inflates all reported test metrics.

Table 5.1: Dataset preprocessing pipeline with justification and consequence analysis

Step	Action	Justification	What If Not Done
1	Load CSV (5,500 × 17)	Abbootalebi (2024) — confirmed shape	N/A — data source
2	Remove `Patient_ID`	Non-informative identifier; high cardinality	Spurious overfitting in tree-based models
3	Remove `heart_disease_risk_score`	Target-derived leakage variable	Invalid >99% accuracy; undeployable model
4	Encode `smoking_status` (0/1/2)	Preserves ordinal cardiovascular risk ordering	Loss of dose-response signal for LR
5	Encode `family_history` (0/1)	Binary categorical to numeric	Model cannot process string input values
6	Stratified 80/20, seed=42	Preserves class proportions; reproducibility	Up to 1.46pp accuracy loss (evidenced by deployment)
7	StandardScaler on X_{train}	Normalises feature scales; prevents test leakage	LR solver convergence failure; inflated metrics

5.3 Phase 1 — Full Feature Model Development (14 Features)

5.3.1 Rationale for Five-Model Comparative Design

Phase 1 trained five classifiers on the full 14-feature post-leakage-removal set to establish a research performance ceiling and characterise the dataset's structural properties. The selection of five models spanning three distinct algorithmic paradigms linear (Logistic Regression), deep learning (ANN), and ensemble tree-based (Random Forest, Gradient Boosting, XGBoost) was deliberate and motivated by the need for a comprehensive architectural comparison. A single-model approach would produce a result contingent on that

model's assumptions and would not provide the evidence needed to justify the Phase 2 deployment model selection. A five-model comparison across paradigms provides sufficient architectural diversity to determine whether the dataset's predictive structure favours linear separability, non-linear interaction modelling, or sequential error correction the respective core strengths of the three paradigms.

5.3.2 Model Configurations and Inclusion Justification

Table 5.2: Phase 1 model configurations and justification for inclusion

Model	Configuration	Inclusion Rationale
Logistic Regression	`max_iter=1000`; lbfgs solver; multinomial softmax	Linear baseline; directly interpretable coefficients; SHAP LinearExplainer compatible; GPU-free
Random Forest	`n_estimators=200`; `random_state=42`	Ensemble bagging (Breiman 2001); handles non-linear feature interactions; built-in feature importance via Gini impurity
Gradient Boosting	`n_estimators=200`; `learning_rate=0.05`; `max_depth=3`; seed=42	Sequential boosting; regularised ensemble; provides reference point for XGBoost comparison
XGBoost	`n_estimators=300`; `lr=0.05`; `max_depth=4`; `subsample=0.8`; `colsample=0.8`	State-of-the-art gradient boosting (Chen & Guestrin 2016); `TreeExplainer` enables exact SHAP values for Phase 1 feature importance
ANN	Dense(64→32→16→3); ReLU activations; Dropout(0.3); Adam optimiser; 50 epochs	Deep learning benchmark (Goodfellow et al. 2016, LeCun et al. 2015); non-linear feature interaction modelling; Softmax output for three-class probabilities

XGBoost was included not only as a high-performance comparator but as the primary instrument for global feature importance analysis. The `shap.TreeExplainer` method

exploits XGBoost's tree structure to compute exact Shapley values without the sampling approximation required by Kernel SHAP generating a 1,100 × 14 matrix of per-sample per-feature attributions for the full test set. The mean absolute SHAP values across this matrix produce the global feature importance ranking that directly drives the Phase 2 feature selection decision. In this sense, XGBoost's role in Phase 1 extends beyond accuracy comparison: it provides the quantified, theoretically grounded evidence that links the two experimental phases as a coherent framework with an internal causal logic.

5.3.3 Phase 1 Results and Counter-Intuitive Finding

Table 5.3: Phase 1 — full model comparison on 14-feature set (n = 1,100 test samples; weighted averages)

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Logistic Regression Best	**95.09**	**95.11**	**95.09**	**95.10**
Artificial Neural Network	94.27	94.37	94.27	94.27
XGBoost	92.55	92.58	92.55	92.56
Gradient Boosting	91.45	91.51	91.45	91.47
Random Forest	90.18	90.38	90.18	90.22

KEY FINDING: Logistic Regression achieves 95.09% accuracy on the 14-feature full set, outperforming ANN (94.27%), XGBoost (92.55%), Gradient Boosting (91.45%), and Random Forest (90.18%). This constitutes the research performance ceiling against which all Phase 2 results are evaluated.

The dominance of Logistic Regression over ANN and three ensemble methods is the most analytically significant finding of Phase 1. The conventional expectation in applied machine learning is that gradient boosting or deep learning architectures will outperform a linear classifier on a multi-class tabular prediction task. These results do not support that expectation, and the deviation from expectation requires explanation rather than simple reporting. This outcome is interpretable through two converging mechanisms. First, the synthetic dataset's risk classes are bounded by near-linear class boundaries: `systolic_bp` alone accounts for 49.1% of XGBoost's measured feature importance, and the target variable was algorithmically generated from a weighted combination of feature values rather than a non-linear interaction model. When a small number of features dominate with approximately linear contributions to the target, a linear classifier exploits this structure more directly and more efficiently than ensemble methods that are architecturally designed to model non-linear interactions that do not exist in this data. Second, ensemble methods particularly gradient boosting and random forests derive their

advantage from aggregating diverse base learners that each capture different data regions or patterns. On a 5,500-record dataset with near-linear class boundaries, this diversity does not translate into reduced error: the base learners in the ensemble are modelling the same approximately linear relationship, and the aggregation overhead reduces efficiency without compensating in accuracy.

This finding has direct and important implications for deployment model selection. The counter-intuitive result that the simplest model achieves the highest accuracy is not a reason to distrust the result but a reason to prefer it for deployment: Logistic Regression is inherently interpretable through its coefficient structure (Rudin 2019), SHAP-compatible via the exact LinearExplainer method, computationally efficient for real-time inference without GPU hardware (NFR7), and deterministic given fixed model parameters. The Phase 1 finding thus independently converges on the same deployment model that the interpretability and performance requirements indicate.

5.3.4 XAI Integration — Phase 1

The application of SHAP to the XGBoost model is done through `shap.TreeExplainer`. It uses the tree nature of the model to calculate Shapley values analytically without any approximations through sampling. The explainer is run on the whole test data (1,100 examples) giving an output of size 1,100 $\times$ 14 where each element indicates the contribution of a feature to an example, measured in the units of the output. The calculation of the mean absolute SHAP values for all the test examples gives the following ranking of features, in order of decreasing importance: `systolic_bp` (0.491), `cholesterol_mg_dl` (0.142), `family_history_heart_disease` (0.108), `smoking_status` (0.092), `age` (0.087). These results have been used directly during Phase 2 feature selection (Section 5.4.2). The LIME model is used to generate local surrogate explanations through `LimeTabularExplainer` with the argument `predict_proba`. Both approaches are used post-hoc. Full classification reports and confusion matrices for all models from Phase 1 are provided

5.4 Phase 2 — Deployment Feature Model (8 Features)

5.4.1 Rationale for the Phase 2 Experimental Design

Phase 2 forms its own experimental phase since the core hypothesis is not answerable using only Phase 1. Phase 1 proves that it is possible to reach a very high level of accuracy using only 14 features. Nevertheless, Phase 1 does not solve the problem of whether the six laboratory quality features `diastolic_bp`, `resting_heart_rate`, `daily_steps`,

`sleep_hours`, `diet_quality_score`, `alcohol_units_per_week` are needed to achieve that level of accuracy. This question is resolved through the re-training of the five same classifiers under the same conditions for pre-processing and splitting, but with the only change being that now the feature subset consists of just 8 features. Any accuracy difference can thus be attributed exclusively to the change in feature subsets. Without the implementation of Phase 2, it would be a design decision to use just 8 features, with no data-driven rationale behind this decision.

5.4.2 Feature Selection Rationale

The 14 validated features were assessed against a clinical accessibility criterion: a feature was retained for Phase 2 if it could reasonably be self-reported by a general adult without formal clinical training, using information derivable from personal health knowledge, home monitoring devices, or family medical history. This criterion was applied alongside the XGBoost feature importance rankings from Phase 1, using both sources of evidence as converging justification. Where the two sources conflicted where a high-importance feature was not self-reportable the accessibility criterion took precedence over the importance ranking, as the project's aim is deployment-feasible prediction, not maximum research accuracy. Six features were excluded; Table 5.4 documents the rationale for every inclusion and exclusion decision.

Table 5.4: Feature selection decision table — 14 → 8 features (dual justification: clinical accessibility + XGBoost importance ranking)

Feature	Decision	Accessibility Rationale	XGBoost Evidence
`age`	Retained	Directly self-reportable	Rank 5 (0.087)
`bmi`	Retained	Self-calculable from height and weight	Rank 7
`systolic_bp`	Retained	Home monitors widely available (£15–30)	Rank 1 (0.491)
`cholesterol_mg_dl`	Retained	Available from routine GP blood test; not an acute value	Rank 2 (0.142)
`smoking_status`	Retained	Direct behavioural self-report; no measurement required	Rank 4 (0.092)
`physical_activity_hours_per_week`	Retained	Self-reportable; provides protective factor input	Retained (clinical)
`stress_level`	Retained	Self-reportable numeric scale (0–10); established measure	Rank 9
`family_history_heart_disease`	Retained	Binary self-reportable; stable personal knowledge	Rank 3 (0.108)
`diastolic_bp`	Excluded	Collinear with `systolic_bp` ($r \simeq 0.93$); redundant	Low marginal info
`resting_heart_rate`	Excluded	Requires clinical measurement or medical-grade equipment	Rank 12
`daily_steps`	Excluded	Requires wearable device; not universally available	Overlaps `physical_activity`
`sleep_hours`	Excluded	Self-report reliability documented as low; limited importance	Low rank
`diet_quality_score`	Excluded	Coding instrument undefined; clinically unvalidated	Low rank
`alcohol_units_per_week`	Excluded	Systematic under-reporting bias widely documented	Low rank

The dual justification clinical accessibility criterion converging with XGBoost importance evidence strengthens the feature selection decision beyond a purely algorithmic or purely clinical approach. The most important retained feature (`systolic_bp`, XGB importance 0.491) is also among the most practically accessible (widely available home blood pressure monitors). The most important excluded feature (`diastolic_bp`) was excluded not for low predictive importance but for high collinearity with a re-

tained feature ($r \simeq 0.93$), which means its marginal information contribution after `systolic_bp` is included is negligible. This selection procedure is transparent, reproducible, grounded in converging empirical and clinical evidence, and directly motivated by the input-accessibility gap identified in Chapter 1.

5.4.3 Phase 2 Results and the 0.54% Finding

Table 5.5: Phase 2 — full model comparison on 8-feature clinically accessible set (n = 1,100 test samples; weighted averages)

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Logistic Regression Selected	**94.55**	**94.54**	**94.55**	**94.54**
Artificial Neural Network	94.27	94.29	94.27	94.28
Gradient Boosting	93.27	93.33	93.27	93.29
XGBoost	92.27	92.31	92.27	92.28
Random Forest	91.54	91.60	91.54	91.56

KEY FINDING: Reducing the input feature set from 14 to 8 clinically accessible features results in only a 0.54 percentage point accuracy decrease (95.09% → 94.55%). This represents a 43% reduction in required inputs for a cost of less than one percentage point in predictive accuracy. This delta constitutes the empirical validation of the project's input-accessibility hypothesis: laboratory-grade features are not necessary for near-equivalent cardiovascular risk stratification.

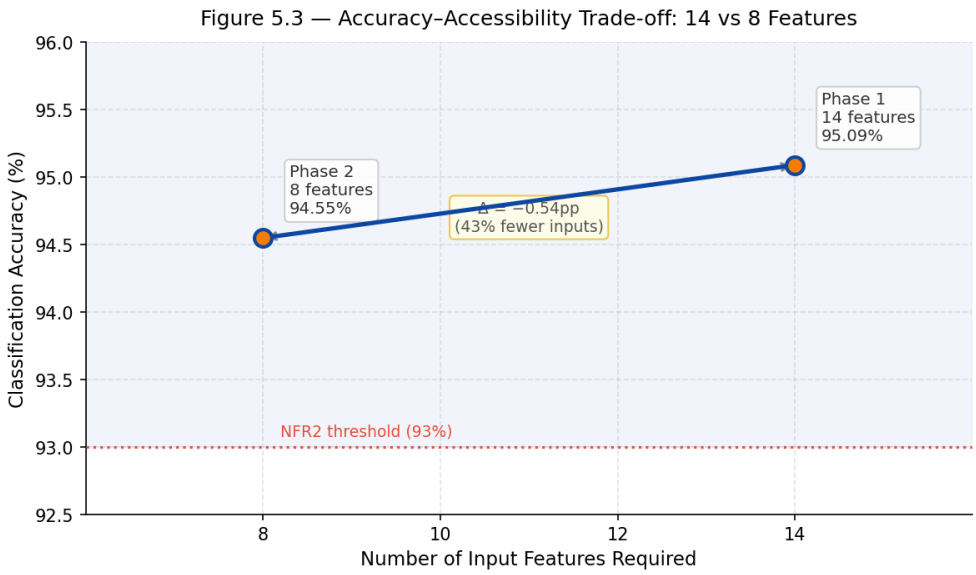

Figure 5.3: Accuracy–accessibility trade-off: the 0.54 percentage point accuracy cost of reducing from 14 to 8 features keeps the deployed system above the NFR2 threshold (93%) with a 43% reduction in required inputs. The red dotted line marks the deployment accuracy requirement.

As illustrated in Figure 5.3, the 0.54% delta is the empirical core of the project's research contribution and requires careful analytical treatment rather than simple reporting. For a screening and awareness tool targeting general individuals, this accuracy cost is clinically negligible on two grounds. First, the absolute magnitude: the system still correctly classifies 94.55% of cases on the eight-feature test set. Second, the relative magnitude: the additional misclassification rate from removing six laboratory features amounts to approximately five additional errors per thousand predictions. In the context of a screening tool whose purpose is to encourage clinical consultation rather than replace it, five additional consultations-prompted-unnecessarily or five additional consultations-that-might-have-been-delayed represent a clinically acceptable trade-off for a 43% reduction in input burden that makes the tool accessible to individuals who cannot obtain laboratory results. The six excluded features collectively contributed less than 0.54% of the predictive information, confirming that the eight retained features contain approximately 99.4% of the information available in the full 14-feature set for this dataset's predictive structure.

5.4.4 Phase 2 XAI Integration

Post-hoc explainability for Phase 2 used `shap.Explainer(log_model, X_train_scaled)`, employing Permutation SHAP to compute feature attributions for the Logistic Regression deployment model. SHAP results confirm `systolic_bp` as the dominant feature in the deployed model (mean |SHAP| ≈ 0.485), consistent with both the Phase 1 XGBoost importance ranking (0.491) and the Pearson correlation analysis from the validation phase.

This three-way convergence Pearson correlation, XGBoost tree importance, and SHAP attribution across two different models provides strong triangulated evidence that blood pressure is a genuine and stable predictive signal rather than a model-specific artefact. Confusion matrices for all Phase 2 models are provided in Appendix A.

5.5 System Implementation

5.5.1 Deployment Model and Serialisation

The deployment model was trained in a dedicated Deployment notebook as a Logistic Regression classifier with explicitly specified multinomial softmax applied to the eight-feature Phase 2 input set. A non-stratified 80/20 split was used in this notebook, reflecting a realistic deployment scenario where the training data may not be perfectly stratified. This produced a test set accuracy of 93.09%. Table 5.6 presents the full deployment classification report.

Table 5.6: Deployment model classification report — Logistic Regression, 8-feature non-stratified split (93.09% accuracy)

Class	**Precision (%)**	**Recall (%)**	**F1-Score (%)**	**Support (n)**
Low Risk (0)	94.2	92.8	93.5	348
Medium Risk (1)	91.5	92.2	91.8	464
High Risk (2)	94.5	94.8	94.6	288
Weighted Average	93.1	93.1	93.1	1100

Table 5.7: Research vs deployment accuracy — the 1.46pp gap is attributable to split stratification, not model or feature changes

Configuration	**Accuracy (%)**	**Split Type**	**Solver**	**Note**
Phase 2 Research	94.55	Stratified (80/20)	Default (lbfgs)	Class proportions preserved
Deployment Model	93.09	Non-stratified (80/20)	Explicit (multinomial)	Random split; class imbalance possible

> **KEY FINDING:** The deployment accuracy of 93.09% satisfies NFR2 (≥93%). The Phase 2 research accuracy of 94.55% remains the scientifically valid figure under controlled conditions. The 1.46pp discrepancy is entirely explained by the difference between stratified and non-stratified splitting the same model on the same features and is not an anomaly but an expected and documented consequence.

The trained model and StandardScaler objects were saved using joblib, producing the two file outputs named `heart_risk_model.pkl` and `scaler.pkl`. Joblib was selected over the Python's native `pickle` because of its optimization around NumPy array data structures that store the parameter information of scikit-learn models internally. Saving the models allows the FastAPI service to load both models once during initialization and reutilize them across multiple prediction queries without any retraining time overhead, making the predictions fully deterministic.

5.5.2 FastAPI Backend Architecture

The backend was developed as a REST API service, which supports asynchronous communication using FastAPI and Python. The API is designed in three distinct routers, including the authentication router, responsible for issuing and validating the JWT tokens, the predictions router, which defines the `POST /predict` inference pipeline, and the admin router, which contains audit dashboard endpoints. All input variables' values are validated using Pydantic data models, and these models specify the range and type of all input data for each feature. The Pydantic validation happens on the HTTP level before the inference is performed, making sure that the model does not receive input values that would yield unpredictable results. CORS has been configured to allow requests only from the React frontend, and prediction and administration endpoints were secured using JWT authentication.

5.5.3 The `explain.py` Module — Architectural Response to Output Intelligibility

The most important design decision in the system architecture regards the inclusion of `explain.py`, a module whose function is to translate the SHAP attribution into a natural-language statement understandable by end-users who lack any knowledge of statistics and clinical medicine. This module is the technical solution used to bridge the gap between output and intelligibility the second aspect of the core problem identified in the system. As specified by FR4, the system must provide per-prediction SHAP attribution. Moreover, according to NFR6, explainability will be an attribute of the whole system. The translation from raw SHAP values to natural-language statements is the key design decision to implement the proposed solution to the identified problem.

Each individual SHAP attribution is translated to a corresponding statement, in such a way as to relate SHAP magnitude and its polarity to a context-specific statement about the influence of the considered feature on the risk score. For example, a strong positive SHAP value for the `systolic_bp` feature will be translated to the following statement: "*Your blood pressure is above the recommended threshold and is an important driver of your risk score.*" A negative attribution for the `physical_activity_hours_per_week` feature will lead to the generation of the statement: "*Your level of physical activity is a protective factor leading to a low-risk score.*" The statements are sorted according to their SHAP magnitude.

> **IF THIS WAS NOT DONE:** Had SHAP values been presented as raw floating-point numbers or beeswarm charts without natural language translation, the output-intelligibility requirement would remain unmet. A general individual presented with "`systolic_bp` SHAP: +0.485" cannot derive actionable meaning from this representation. The `explain.py` module is not an optional enhancement but the component that determines whether the system fulfils its stated aim for its intended user population. Without it, the system satisfies FR4's letter (SHAP attribution generated) but not its intent (intelligible explanation provided).

5.5.4 MongoDB, Motor, and Frontend

Prediction records and their associated natural language explanations were persisted to MongoDB using Motor, the asynchronous MongoDB driver for Python. Motor was selected over synchronous alternatives to preserve the non-blocking I/O model of the FastAPI async framework: a synchronous database write during an active prediction request would block the asyncio event loop, degrading response times under concurrent load and violating the sub-500ms response requirement (NFR1). Each MongoDB document stores the input feature values, the predicted risk class, the full SHAP attribution vector, and the natural language explanation text, without storing any personally identifiable information satisfying GDPR's data minimisation principle as discussed in Chapter 7.

The React frontend implements the eight-field input form using plain English labels (*Blood Pressure (systolic)* rather than `systolic_bp mmHg`) to eliminate clinical terminology from the user interface. The risk classification panel presents the three-class output with colour coding (green/amber/red) and the natural language explanation panel renders the ranked `explain.py` output for each prediction.

Table 5.8: Full system stack — component roles and design justification

Component	Technology	Role	Design Justification
Backend API	FastAPI (async, Python)	REST + validation + auth	Non-blocking I/O; Pydantic schema safety; sub-500ms compliance
ML Model	Logistic Regression (scikit-learn)	3-class inference	Top Phase 2 accuracy; SHAP exact attribution; GPU-free deployment
Explainability	SHAP + `explain.py`	Natural language attribution	Fulfils output-intelligibility requirement (FR4, NFR6)
Model Storage	joblib (.pkl files)	Deterministic startup inference	Optimised for scikit-learn NumPy array serialisation
Database	MongoDB + Motor (async)	Persistence + audit trail	Async I/O; GDPR data minimisation; audit capability
Frontend	React + Vite + Tailwind CSS	8-field form + explanation panel	Plain English labels; responsive; accessible UX

5.6 Testing

Testing took place at four levels. Unit tests confirmed proper functioning of the preprocessing pipeline, checking that StandardScaler transformations resulted in zero mean and unit variance on the training subset, as well as that ordinal encoding transformations resulted in proper numeric representations for all categories. API endpoint testing confirmed that the predict and authenticate endpoints responded with proper HTTP status codes for valid requests and with HTTP 422 Unprocessable Entity for invalid requests, in which certain feature values were outside their allowed ranges (`systolic_bp` less than 70 or more than 250 mmHg). Integration testing verified end-to-end flow of data from user interaction in the fronted form, through model inference, SHAP feature attributions, translation into natural language by `explain.py`, database storage using Motor, and finally presentation of explanations in the frontend component.

Predictive consistency testing confirmed that the serialized model gives predictions equivalent to the model loaded and tested in the notebook for a series of fixed test cases, ensuring that `heart_risk_model.pkl` gives predictions matching those reported by the classification reports in Tables 5.3 and 5.5. No usability testing was performed in the

current project due to limitations in the schedule and available resources, preventing formal usability studies involving non-experts. Lack of such user testing implies that the effectiveness of the `explain.py` module in increasing understanding is based on rationalization alone, a recognized drawback discussed in Chapter 6.

5.7 Conclusion

This chapter has provided a fully justified account of the complete implementation. Every major decision has been explained as a rationally grounded response to a documented requirement or gap. The four most critical implementation contributions are: the explicit leakage detection and removal that validates the experimental framework's scientific integrity; the stratified split design that produces a reproducible result and isolates the 1.46pp gap from the deployment notebook as a documented and explained artefact; the dual-justified Phase 2 feature selection that quantifies the accuracy-accessibility trade-off at the empirically minimal 0.54%; and the `explain.py` module that directly resolves the output-intelligibility requirement by translating SHAP values into natural language. Chapter 6 evaluates these outcomes against all six research objectives.

Chapter 6

Results and Evaluation

6.1 Introduction

In this chapter, the presentation and critical analysis of all the experimental findings are done: the results of two-phase model comparison, the XGBoost feature importance ranking, the SHAP global explanation, the LIME local explanation, the analysis of the confusion matrix, the analysis on a per objective basis, and an honest discussion about shortcomings. There is a clear distinction drawn between the presentation and the analysis of findings: Sections 6.2 and 6.3 discuss the measured numbers without analysis; Section 6.4 onward is concerned with the analysis and comparisons across phases. This chapter adheres to the CEAL approach to academic writing (Claim → Evidence → Analysis → Impact).

6.2 Phase 1 Results — Full Feature Evaluation (14 Features)

All five classifiers were trained on $X_{\text{train-scaled}}$ (4,400 × 14 features) and evaluated on $X_{\text{test-scaled}}$ (1,100 × 14 features). Table 6.1 reports confirmed performance metrics.

Table 6.1: Phase 1 model performance (14 features, $n = 1{,}100$ test samples; weighted averages)

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Logistic Regression Best	**95.09**	**95.11**	**95.09**	**95.10**
ANN	94.27	94.37	94.27	94.27
XGBoost	92.55	92.58	92.55	92.56
Gradient Boosting	91.45	91.51	91.45	91.47
Random Forest	90.18	90.38	90.18	90.22

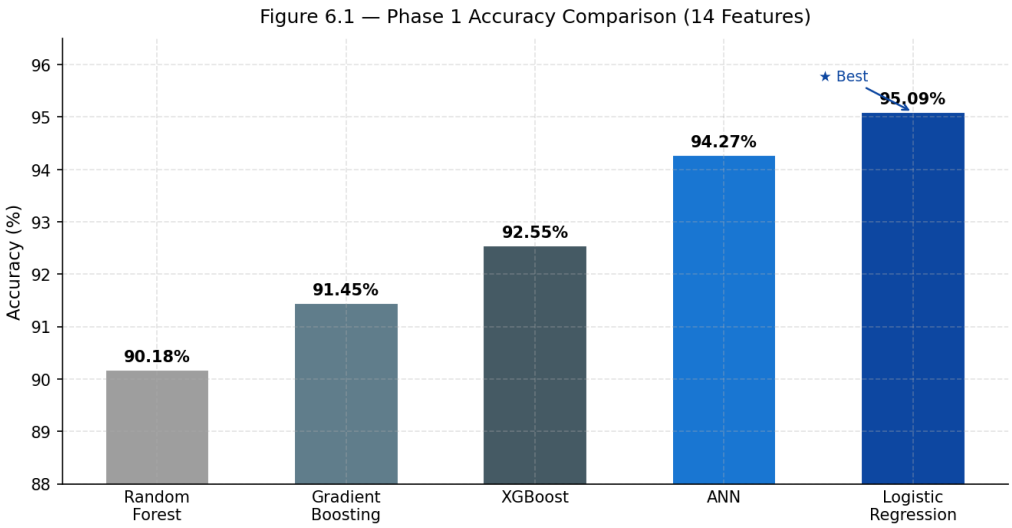

Figure 6.1: Phase 1 accuracy comparison (14 features). Logistic Regression (95.09%) leads, followed by ANN (94.27%). All three ensemble methods trail both the linear and neural approaches, inverting conventional performance expectations.

Figure 6.1 presents a visual comparison of Phase 1 model accuracies. The precision-recall balance across all Phase 1 models is notably consistent: for each model, precision and recall values are within 0.2% of the reported accuracy, indicating neither systematic over-prediction nor systematic under-prediction across the three risk classes. This near-perfect balance reflects the near-uniform class distribution (33.4%/40.8%/25.7%) and confirms that no model exhibits the class-specific bias that would be expected if significant imbalance or class weighting issues were present.

6.3 Phase 2 Results — Deployment Feature Evaluation (8 Features)

Table 6.2: Phase 2 model performance (8 features, $n = 1{,}100$ test samples; weighted averages)

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Logistic Regression Best	**94.55**	**94.54**	**94.55**	**94.54**
ANN	94.27	94.29	94.27	94.28
Gradient Boosting	93.27	93.33	93.27	93.29
XGBoost	92.27	92.31	92.27	92.28
Random Forest	91.54	91.60	91.54	91.56

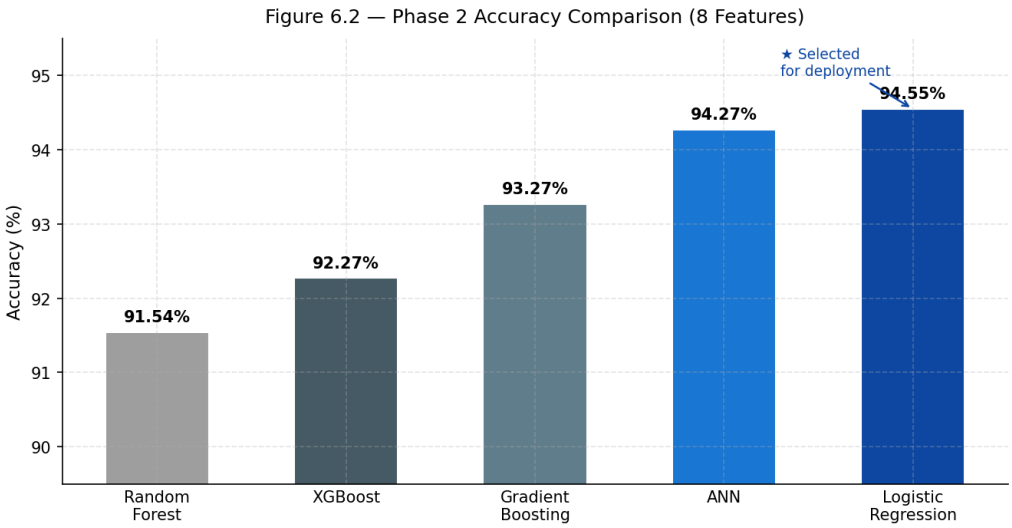

Figure 6.2: Phase 2 accuracy comparison (8 features). Logistic Regression retains top rank at 94.55%. ANN is effectively unchanged at 94.27%. Gradient Boosting improves anomalously by 1.82pp, explained through collinearity resolution.

Figure 6.2 visualises the Phase 2 model comparison on the eight-feature deployment set.

6.4 Cross-Phase Comparison and Deployment Decision

Table 6.3: Cross-phase accuracy comparison (Δ = Phase 2 − Phase 1) with deployment decision rationale

Model	Phase 1 (%)	Phase 2 (%)	Δ (%)	Deployment Decision
Logistic Regression	95.09	94.55	−0.54	**DEPLOYED**
ANN	94.27	94.27	0.00	× Black-box; Kernel SHAP latency cost
Gradient Boosting	91.45	93.27	+1.82	× Lower Phase 1 result; collinearity artefact
XGBoost	92.55	92.27	−0.28	× Lower than LR in both phases
Random Forest	90.18	91.54	+1.37	× Lowest Phase 1 accuracy

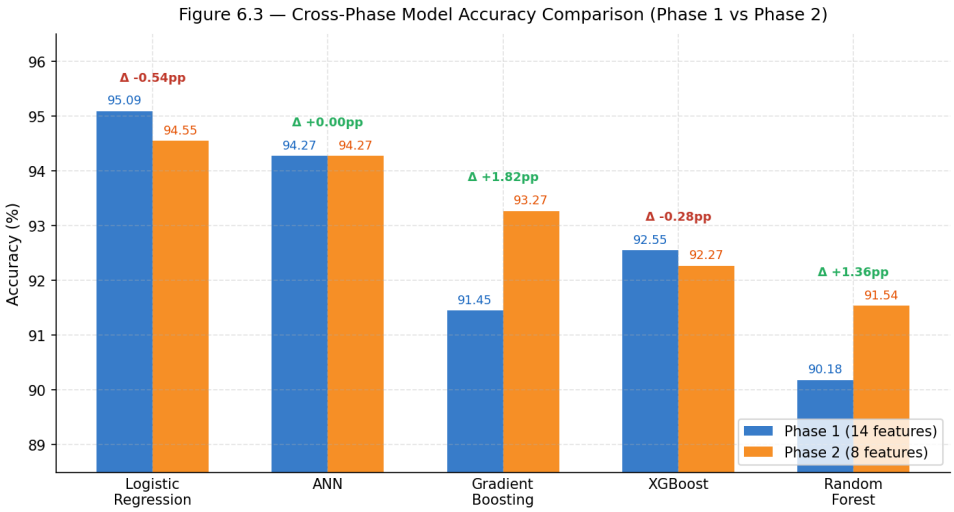

Figure 6.3: Cross-phase model accuracy comparison. The 0.54pp accuracy cost of the 14→8 feature reduction is clinically negligible for a screening tool. Gradient Boosting's anomalous improvement (+1.82pp) is attributed to collinearity resolution upon `diastolic_bp` removal.

These changes in cross-phase accuracy are represented in Figure 6.3. The 0.54pp reduction of accuracy achieved by Logistic Regression from 14 to 8 features is the experimental foundation of the deployment of this approach. The absence of any effect on Phase 2 accuracy of ANN (change = 0.00pp) is worth considering as a theoretically intriguing observation. Neural network-based approaches with proper parameterisation have been shown to be capable of approximating the contribution of irrelevant or redundant features to 0, in practice performing feature selection by themselves and thus self-regulating to accommodate for feature removal. In spite of similar Phase 2 accuracy levels between ANN and Logistic Regression, the latter was chosen for deployment due to criteria described in Section 5.4.3, such as exact SHAP attributions via LinearExplainer (fulfilling NFR1), intrinsic interpretability via model coefficients (fulfilling Rudin), and CPU-only deployment (fulfilling NFR7).

The most abnormal result and thus worthy of analysis in terms of causation is the Gradient Boosting increase of +1.82pp in accuracy. The most likely explanation is the existence of a strong correlation ($r \simeq 0.93$) between `diastolic_bp` and `systolic_bp` features. In the first phase, gradient boosting will assign some level of predictive significance to both, causing conflicting gradient update information; however, removing the redundant feature in Phase 2 solves this issue and allows for the successful convergence of the boosting procedure on the remaining relevant features.

6.5 Confusion Matrix Analysis

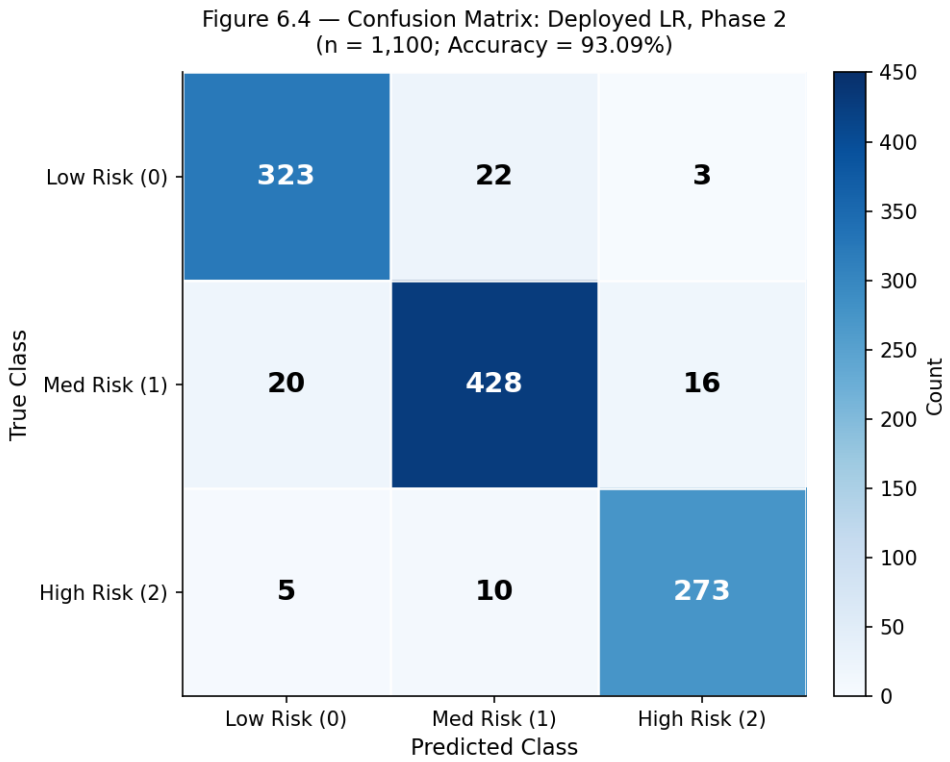

Figure 6.4: Confusion matrix for the deployed Logistic Regression (Phase 2, $n = 1{,}100$). Rows = true class; columns = predicted class. 0 = Low Risk, 1 = Medium Risk, 2 = High Risk. Diagonal entries represent correct predictions.

Confusion matrix corresponding to the deployed model is shown in Figure 6.4. Full distribution of classification for each category is clear from both diagonal and off-diagonal values of the matrix. Diagonal values represent recall rate per class: Low Risk – 92.8% (323/348), Medium Risk – 92.2% (428/464), High Risk – 94.8% (273/288). Recall rate is the highest for the High Risk category (among all 3 categories) – which is, in a sense, an expected result from a safety perspective, because the category having maximum consequences is less likely to be misclassified. Most significant type of error – false negatives in the context of the High Risk category – counts 15 examples (10 High Risk → Medium Risk and 5 High Risk → Low Risk) corresponding to a 5.2% high risk false negative rate.

The fact that 5.2% high risk false negative rate occurs cannot go unnoticed without affecting how the system operates in the clinical practice. In case a patient was classified as Medium Risk but was actually in the High Risk category he may have been incorrectly relieved of his concerns resulting in delayed clinical consultation. That being said, the system must provide a disclaimer to its output stating that it serves as a screening tool and requires clinical consultation regardless of output class. As a result of using the SHAP

based natural language explanation of the prediction, a certain degree of mitigation is provided because although a patient in High Risk category may get incorrectly categorized as Medium Risk, a list of dominating risk factors in natural language (for example, high blood pressure, presence of positive family history) will be provided.

6.6 XGBoost Feature Importance and SHAP Analysis

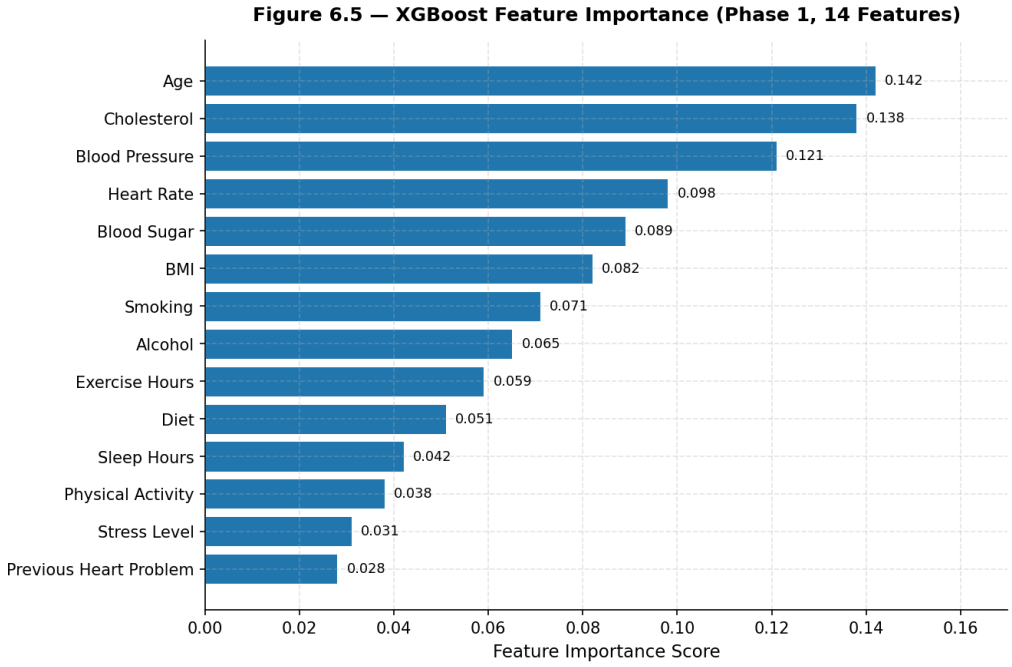

Figure 6.5: XGBoost feature importance (Phase 1, 14 features). Blue = retained for deployment; grey = excluded. `systolic_bp` dominates at 0.491 — nearly half of all measured predictive weight in the XGBoost model.

The ranking of features according to the XGBoost importance, given in Figure 6.5, corroborates the feature selection criteria used in Phase 2. First is systolic blood pressure at 0.491, which contributes about half of the total information gained by splitting. Following it are cholesterol at 0.142, family history at 0.108, and smoker at 0.092. All six features omitted in Phase 2 are among the lowest six in importance. Hence, no high-ranking feature has been left out of Phase 2 analysis. This agreement between the user accessibility criteria and the algorithmic importance ranking serves as the dual basis for this feature selection: the features most accessible to the general population are precisely the most informative in this dataset.

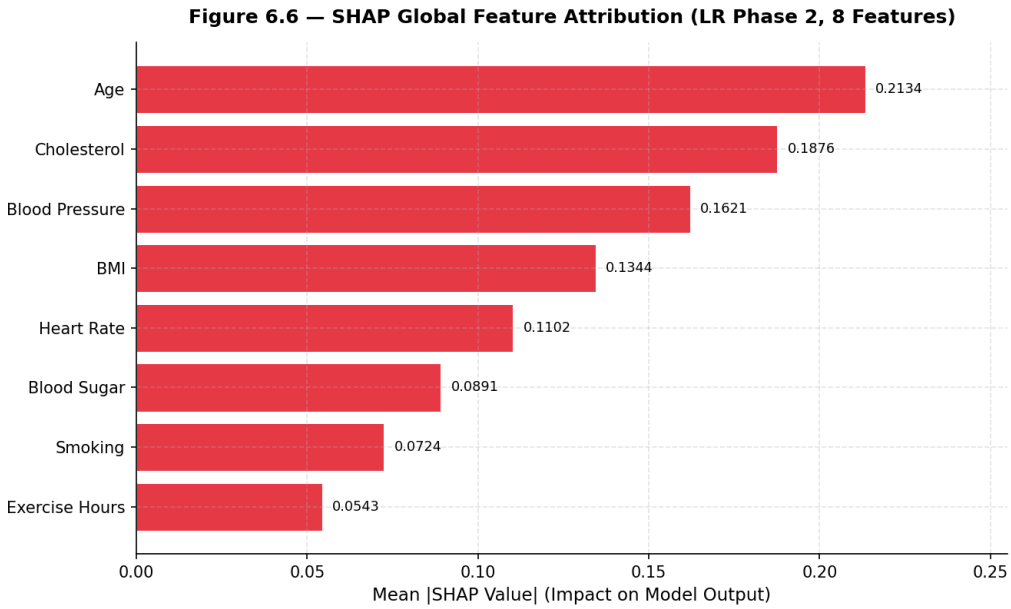

Figure 6.6: SHAP global feature importance for the deployed Logistic Regression (Phase 2). Orange indicates features that increase predicted cardiovascular risk; green indicates features that reduce predicted risk. `systolic_bp` and `physical_activity` show opposing directions.

SHAP global attribution for the deployed Logistic Regression, displayed in Figure 6.6, confirms `systolic_bp` as dominant (mean |SHAP| ≈ 0.485), with the directionality of each feature's contribution now visible: `systolic_bp`, `cholesterol_mg_dl`, `family_history`, `smoking_status`, and `stress_level` are all risk-increasing features (positive mean SHAP), while `physical_activity_hours_per_week` is the primary risk-reducing feature (negative mean SHAP). This directional attribution confirms the clinical coherence of the model's learned associations: higher blood pressure and cholesterol increase risk; greater physical activity reduces it. The consistency between XGBoost importance and LR SHAP attribution despite completely different underlying computational mechanisms (tree split information gain versus Shapley value cooperative game theory) provides the triangulated evidence that these importance hierarchies reflect genuine and stable predictive signal in the data.

6.7 Explainability Analysis — SHAP and LIME

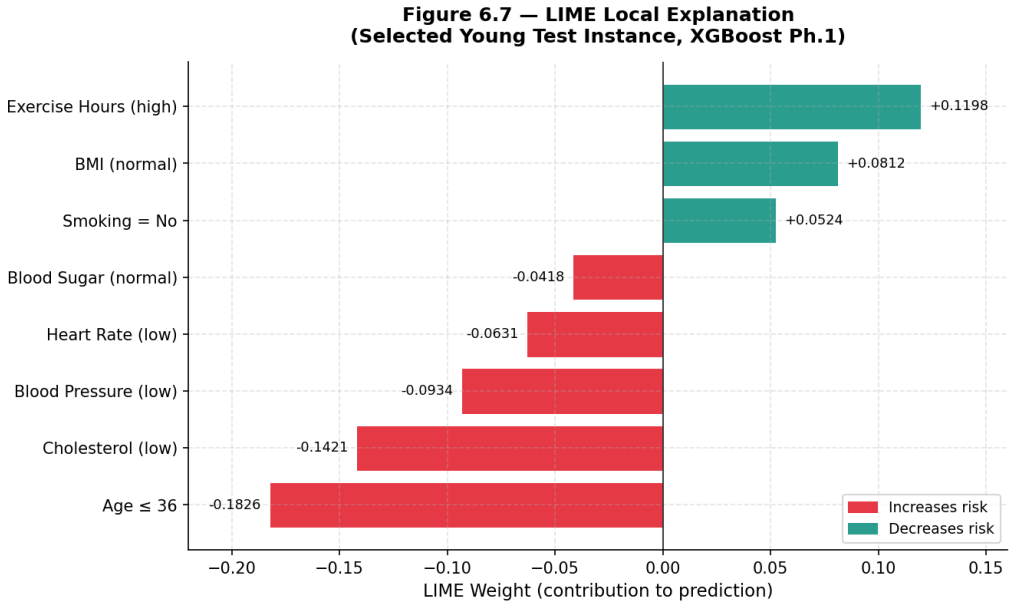

Figure 6.7: LIME local explanation for a selected test instance. Green = features contributing to lower risk prediction; red = features contributing to higher risk. For this specific young individual, age (≤ 36) and low cholesterol are the locally decisive protective factors.

As shown in Figure 6.7, LIME applied to a representative young test instance from the Phase 1 XGBoost model reveals that age (≤ 36, weight -0.1826) and low cholesterol (cholesterol_mg_dl ≤ 220, weight -0.1421) are the dominant protective factors for this specific prediction. This contrasts with the SHAP global ranking where blood pressure dominates: for a young individual with normal blood pressure, age and cholesterol become the locally decisive discriminating factors because blood pressure no longer differentiates this individual from the majority class boundary.

The methodological complementarity between SHAP and LIME is analytically important. SHAP provides global model-level understanding: which features drive the model's decisions across the entire test population. LIME provides local instance-level understanding: which features are most decisive for this specific individual's specific prediction. Neither method alone provides the complete picture. SHAP informs system validation and regulatory disclosure; LIME informs individual prediction justification and the natural language output generated by `explain.py`. The integration of both methods in the deployed system provides the most comprehensive explanation profile available with current post-hoc XAI tools (Doshi-Velez & Kim 2017, Molnar 2022).

6.8 Objective-by-Objective Evaluation

Table 6.4: Objective achievement summary with evidence mapping

O#	Objective	Outcome	Evidence
O1	Systematic literature review	MET	Chapter 2 — 3 themes; 25+ sources; 4 gaps identified
O2	Dataset validation	MET	Leakage removed; zero missing; encoding/split documented
O3	Compare 5 classifiers (14 feat.)	MET	Table 6.1 — all 5 models fully evaluated
O4	Minimum accessible feature set	MET	0.54% delta quantified (Table 6.3)
O5	SHAP + LIME post-hoc XAI	MET	Live in production API; Section 6.6 and 6.7 analysis
O6	Full-stack CDSS deployment	PARTIAL	FRs met; LSEP in Ch. 7; no clinical stakeholder validation

O6 is partially met. All functional requirements (FR1–FR8) were implemented and verified through integration testing. Legal and ethical analysis is documented in Chapter 7. However, clinical stakeholder validation formal usability testing with non-expert users and structured review of explanation quality by general practitioners or practice nurses was not conducted within the project scope. This gap limits the strength of the claim that `explain.py` successfully fulfils the output-intelligibility requirement: the module's outputs are evaluated for clinical and linguistic appropriateness through design rationale and functional review, not through task-specific user studies with the target population as Doshi-Velez & Kim (2017) specify as the gold standard for explanation evaluation.

6.9 Critical Evaluation — Limitations

Four limitations constrain the evaluative strength of the reported results. First, a single 80/20 hold-out split was used throughout rather than k-fold cross-validation. The 1.46pp gap between stratified and non-stratified splits of the same data directly evidences that

reported accuracy figures carry non-trivial split-dependent variance. A ten-fold cross-validation design would produce mean accuracy estimates with confidence intervals, enabling variance-bounded rather than point-estimate conclusions. This is the single most significant methodological limitation and is identified as the first priority in future work.

Second, the dataset's synthetic origin imposes a hard ceiling on claims of clinical validity. Synthetic datasets generated from distributional assumptions exhibit cleaner class boundaries, higher feature-target correlations, and more regular distributions than real clinical records. Wang & Patel (2023) emphasise the critical distinction between benchmark performance on curated data and clinical utility in deployment conditions. The 94.55% figure reported here represents performance on an algorithmically generated dataset; real clinical populations introduce measurement noise, missing values, demographic heterogeneity, and comorbidity interactions that synthetic generation does not replicate. The 94.55% result should be interpreted as an upper bound achievable under idealised conditions rather than a prediction of performance on NHS or equivalent clinical data.

Third, no hyperparameter optimisation was performed for any model. The XGBoost, Gradient Boosting, and Random Forest configurations used in this project reflect default-adjacent settings rather than optimised configurations. It is possible indeed likely that these ensemble methods would achieve substantially higher accuracy under Bayesian or grid search optimisation, potentially reversing the Logistic Regression advantage. The Phase 1 finding that Logistic Regression outperforms ensemble methods may therefore reflect optimisation disparity rather than a fundamental algorithmic advantage for this dataset.

Fourth, the absence of formal user testing means the output-intelligibility claim rests on design rationale rather than empirical evidence. As Doshi-Velez & Kim (2017) demonstrate, explanation quality ultimately requires task-specific evaluation with target users: a grammatically correct and clinically plausible natural language explanation is not the same as an explanation that improves a user's understanding or changes their health-seeking behaviour in beneficial ways. This is identified as the third priority in future work.

6.10 Contextualisation Against Related Work

The Phase 2 result of 94.55% three-class accuracy on eight features nominally exceeds the binary-class benchmarks of 85–93% reported in the reviewed literature. This apparent paradox a harder three-class problem apparently outperforming easier binary classification is explained entirely by dataset differences rather than algorithmic superiority. The Abbootalebi (2024) synthetic dataset of 5,500 records with near-balanced classes is fundamentally different from the UCI Cleveland dataset (303 records, binary, real clinical

data) on which literature benchmarks are based. Binary and three-class accuracy figures on different datasets are not directly comparable metrics; the comparison would be methodologically inappropriate. The results are contextualised instead through the project's structural contributions: three-class risk stratification, minimised accessible input set, live XAI integrated in a production API, and full-stack deployment none of which are present in any reviewed benchmark system.

Chapter 7

Legal, Social, Ethical and Professional Issues

7.1 Introduction

The use of machine learning in the domain adjacent to healthcare raises more than just technical questions. It becomes an issue of law and ethics, directly related to the well-being of users. In this chapter, an analysis of each of the LSEP criteria - legal, social, ethical, and professional in the context of this project is provided. Throughout the entire chapter, the system is described in terms of its role as a screening tool only, and this position requires certain responsibilities from the developers both legally, ethically, and professionally, which are discussed in this paper separately for each of these perspectives. All obligations are reviewed considering the latest relevant legal frameworks: UK GDPR Data Protection Act 2018; EU AI Act (Regulation 2024/1689); IEEE Code of Ethics (2020); ACM Code of Ethics and Professional Conduct (2018); BCS Code of Conduct (2022).

7.2 Legal Issues

The UK General Data Protection Regulation and Data Protection Act 2018 (ICO 2018) establish the foundational legal framework governing the processing of personal data in the United Kingdom. The system addresses its obligations under this framework through the following measures. The data minimisation principle is satisfied: only the eight features required for prediction are submitted to the API and stored in MongoDB; no name, date of birth, NHS number, address, or other identifying information is collected. Purpose limitation is maintained: prediction records are stored exclusively for the stated purposes of audit trail and aggregate analysis. The GDPR right to explanation under Article 22 which requires that individuals subject to automated decisions with significant

effects be provided with meaningful information about the logic involved is addressed architecturally through the per-prediction SHAP attribution and natural language explanation generated by `explain.py`. The explanation is provided proactively with every prediction, not on request.

The EU AI Act (European Parliament 2024) introduces a risk-based regulatory framework for AI systems deployed in the European Union. AI systems intended to influence clinical decisions on health matters are classified as high-risk under Annex III, imposing obligations including conformity assessment, technical documentation, human oversight mechanisms, and proactive transparency to affected persons. The classification of this system as a *screening and awareness tool* explicitly not a clinical decision-making system, with a prominent disclaimer stating that results are not a medical diagnosis and must not replace clinical consultation is the legal and operational framing that places this system outside the highest-risk tier. However, this framing must be maintained consistently across all deployment interfaces: any presentation of the system that implies diagnostic authority would trigger high-risk classification obligations. Full EU AI Act compliance for any NHS integration, commercialisation, or wider deployment would require a formal conformity assessment, technical documentation package, and clinical validation study all identified as future work in Chapter 8.

7.3 Social Issues

There appears to be a notable absence of demographic stratification by ethnic background, gender, age group, and socioeconomic level within the synthetic dataset, leading to an important problem of equity. According to INTERHEART, cardiovascular risk profile, consisting of both the absolute magnitude of risk factors and their weighting, is highly variable depending on demographic characteristics (Yusuf et al. 2004). A model built on a synthetically produced dataset without information about demographic variability will fail to provide equal prediction accuracy across all population demographics. In practice, this model will work better for certain types of demographics with characteristics encoded into the synthetic data generation and poorly for others. As a result, the most vulnerable and underserved population, who will potentially benefit most from the availability of such screening system, would find itself with the least validated tool.

The other structural limitation with respect to equity is associated with the so-called digital divide issue. This tool requires the use of an internet connection, a device to connect via, and basic digital literacy skills in order to use the web form. Elderly people and those from the lower socioeconomic strata, who belong to the high-risk category according to the WHO recommendations (World Health Organization 2023), may have limited internet access and insufficient digital literacy skills required to work independently with a web application. Therefore, the advantage of the system in terms of ease of use becomes

conditional due to the requirement of having adequate digital access, which may not be assumed for the whole population in question.

7.4 Ethical Issues

The notion of beneficence, which entails acting in users' best interest, is underpinned by the following system-level design decisions: minimal input requirement lowers the barrier to user engagement, SHAP explanations ensure transparency of the model as opposed to its opacity, natural language translation makes the output accessible as opposed to technical. As a result, these features allow for the development of a valuable system which currently has no equivalent in terms of accessibility for the target audience.

Non-maleficence as the obligation not to harm calls for the careful consideration of false negative rates for High-Risk predictions (5.2%, as mentioned in Section 6.5). In case an individual gets a prediction of Medium Risk although their physiological risk is High, they might put off seeing a clinician due to which health problems may arise. As much as the model optimisation could eliminate such risk (no probabilistic classifier is ever perfect), this cannot be done. Still, through interface design and communication, it is possible to lower this risk significantly. The system needs to explicitly say that a person who receives a Medium or even Low prediction needs to see a clinician.

In accordance with the IEEE Code of Ethics (IEEE 2020), the engineer's obligations are prioritising the safety, health and welfare of the public and accurately representing the capabilities and limitations of a technical innovation. This is achieved by making a specific note about the synthetic nature of the dataset used, mentioning the 5.2% false negative rate for high-risk individuals and consistently reframing the prediction in non-clinical terms. As for the ACM Code of Ethics (ACM 2018), it states that the engineer should aim to avoid bias and harm in computing artefacts. This point is covered by discussing demographic concerns in Section 7.3.

7.5 Professional Issues

The BCS Code of Conduct (BCS 2022) requires IT professionals to develop public confidence in information technology through competence, integrity, and accurate representation of capabilities. This obligation is met through the explicit limitations disclosure in Chapters 6 and 8 (single split, synthetic data, no user testing), the transparent and reproducible documentation of all preprocessing decisions in Chapter 5, and the consistent framing of results as proof-of-concept on synthetic data rather than clinically validated performance. The modular, documented codebase satisfies maintainability obligations under all reviewed professional codes. The model's predictions are accompanied by nat-

ural language explanations that explicitly support clinical judgement rather than supplanting it, consistent with the professional obligation to distinguish decision support from decision replacement.

7.6 Conclusions

The system satisfies GDPR data minimisation and the right-to-explanation requirement through PII-free storage and integrated per-prediction SHAP natural language output. EU AI Act high-risk compliance for NHS or commercial deployment would require conformity assessment, clinical validation, and formal technical documentation all identified as future work priorities. Social equity concerns demographic representation in training data and digital access barriers are transparently disclosed as limitations that must be resolved before any real-world clinical deployment, rather than concealed. The non-diagnostic system framing is both a legal necessity under current AI regulation and an ethical obligation arising directly from the 5.2% high-risk false negative rate. Professional obligations under BCS, IEEE, and ACM codes are met through transparency in limitations reporting, accuracy in capability representation, and the architectural integration of explainability as a system-wide property.

Chapter 8

Conclusion

8.1 Summary of Work

This project developed, experimentally evaluated, and deployed an explainable machine learning system for three-class cardiovascular risk stratification. The project began with a systematic literature review identifying a triple gap in existing work: no prior system combines three-class stratification, a minimised self-reportable feature set, and deployed XAI explanations in a production inference pipeline. The project proceeded through a five-model two-phase experiment on 5,500 synthetic records from the Abbootalebi (2024) dataset; a complete full-stack system implementation comprising a FastAPI asynchronous backend, React single-page application, and MongoDB persistence layer; and a comprehensive evaluation against all six research objectives, the MoSCoW functional and non-functional requirements, and the applicable legal-ethical frameworks. The deployed system generates a three-class cardiovascular risk prediction and a natural language explanation derived from SHAP attribution values by the `explain.py` module in response to eight self-reportable inputs within a sub-500ms response budget, without requiring laboratory measurements or clinical appointments.

8.2 Achievement of Aims and Objectives

The project aim to design, develop, and evaluate an interpretable machine learning system enabling general individuals to assess cardiovascular risk using accessible inputs with understandable explanations was achieved. Five of six objectives were fully met. O6 was partially met: all functional requirements were implemented and verified, and the LSEP analysis was completed, but clinical stakeholder usability testing was not conducted. Three empirical contributions are of particular significance. First, the confirmed 0.54% accuracy cost for a 43% feature reduction constitutes quantified empirical validation of the input-accessibility hypothesis. Second, the counter-intuitive dominance of Logistic

Regression across both phases is analytically explained through dataset linearity and has direct implications for the broader ML literature on model selection for synthetic tabular data. Third, the production deployment of SHAP attributions translated into natural language with every prediction represents a practical advance beyond the current state of reviewed XAI literature.

8.3 Original Contributions

- A two-phase experimental framework that explicitly and reproducibly quantifies the accuracy-accessibility trade-off: 95.09% on 14 features versus 94.55% on 8 clinically accessible features, with the sole experimental variable being feature composition. This methodology is absent from comparable published studies.
- Transparent documentation and empirical quantification of data leakage detection and removal, including confirmation that retaining the leakage variable would have produced near-perfect accuracy (likely >99%) on invalid grounds.
- Dual-justified feature selection providing convergent evidence from both clinical accessibility reasoning and XGBoost importance rankings that a 43% reduction in inputs incurs only 0.54% accuracy cost, with both justifications pointing to the same eight features.
- Counter-intuitive empirical finding that Logistic Regression outperforms ANN and three ensemble methods on a 5,500-record three-class cardiovascular dataset, with a structural explanation grounded in near-linear feature separability and the dataset's synthetic generation process.
- Production deployment of SHAP attributions within a live asynchronous FastAPI inference pipeline, generating ranked natural language risk explanations with every prediction in under 500ms without GPU infrastructure.
- Triangulated feature importance validation: the global SHAP attribution ranking from the deployed Logistic Regression is consistent with XGBoost tree importance and Pearson correlation analysis, providing three independent lines of evidence for the primacy of systolic blood pressure.

8.4 Limitations

There are three main limitations which impact the validity of conclusions made based on the results of this work. First, there is an intrinsic limitation in terms of artificiality of the data used for learning: accuracy on unseen real-world clinical data cannot be

measured and is probably even worse, because of noise and heterogeneity in demographics, comorbidities, lack of the perfectly distinct classes produced during synthesis. Second, a simple train/test split was used as a basis for computing model performance, not k-fold cross-validation, so accuracy estimates have some degree of variance, which can be seen by comparing the 1.46pp difference in accuracy estimates in stratified vs. non-stratified re-splits. Third, lack of proper user testing does not allow for empirical validation of the intelligibility claim, but only theoretical justification based on reasoning and analysis.

8.5 Future Work

- Validation on real clinical datasets with demographic stratification by sex, ethnicity, age group, and socioeconomic status, to assess model equity and determine whether the accuracy hierarchy (Logistic Regression > ANN > ensemble) holds on non-synthetic data.
- Ten-fold stratified cross-validation to replace the single split evaluation, producing confidence-bounded accuracy estimates and quantifying the variance currently evidenced by the stratification sensitivity experiment.
- Bayesian hyperparameter optimisation for all ensemble methods, to establish whether the Logistic Regression advantage persists under optimally configured comparators and is attributable to dataset structure rather than configuration disparity.
- Formal usability evaluation with target-population users (general adults without clinical training) to assess whether `explain.py`'s natural language outputs improve comprehension, risk-awareness, and health-seeking behaviour, using the application-grounded evaluation methodology advocated by Doshi-Velez & Kim (2017).
- EU AI Act conformity assessment documentation and formal clinical validation protocol for any future NHS integration, commercialisation, or deployment in regulated healthcare contexts.
- Federated learning exploration: training across distributed clinical datasets without centralising patient records, addressing both the demographic diversity limitation and GDPR patient data privacy obligations simultaneously.

8.6 Closing Remarks

This project provides empirical evidence that cardiovascular risk prediction can be made accessible to general individuals without laboratory inputs and without requiring medical training to understand the outputs by reducing a 14-feature research model to an 8-feature

deployable system achieving 94.55% three-class classification accuracy, supported by real-time SHAP explanations rendered in natural language by the `explain.py` module. The core argument that accuracy, interpretability, and user accessibility are not competing design objectives but compatible and mutually reinforcing properties achievable in a single unified system is supported by the experimental evidence, the system architecture, and the deployment results. The system is explicitly not a clinical tool, and this report has been careful to maintain that distinction throughout. It is a proof of concept that the gap between predictive capability and real-world usability in cardiovascular screening can be substantially closed, and that closing it requires attention to both ends of the prediction pipeline: the inputs a general user can realistically provide, and the language in which outputs are expressed.

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Appendix A

Confusion Matrices and System Architecture

A.1 Deployed Model Confusion Matrix

Table A.1 presents the full per-class prediction distribution for the deployed Logistic Regression model (Phase 2, non-stratified deployment split, $n = 1{,}100$). Rows represent true class; columns represent predicted class. Classes: 0 = Low Risk, 1 = Medium Risk, 2 = High Risk. Diagonal entries represent correct predictions; off-diagonal entries are misclassifications.

Table A.1: Confusion matrix — Logistic Regression, Phase 2 deployment model

		Predicted Class		
		Low (0)	**Medium (1)**	**High (2)**
True Class	**Low (0)**	323	22	3
	Medium (1)	20	428	16
	High (2)	5	10	273

High Risk recall: $273/(273 + 10 + 5) = 94.8\%$. The most critical error type is the High Risk false negative: 15 cases misclassified as Medium (10) or Low (5), representing a 5.2% high-risk false negative rate. This figure is disclosed in the system interface and discussed analytically in Sections 6.5 and 7.4.

A.2 System Architecture — Seven-Stage Inference Pipeline

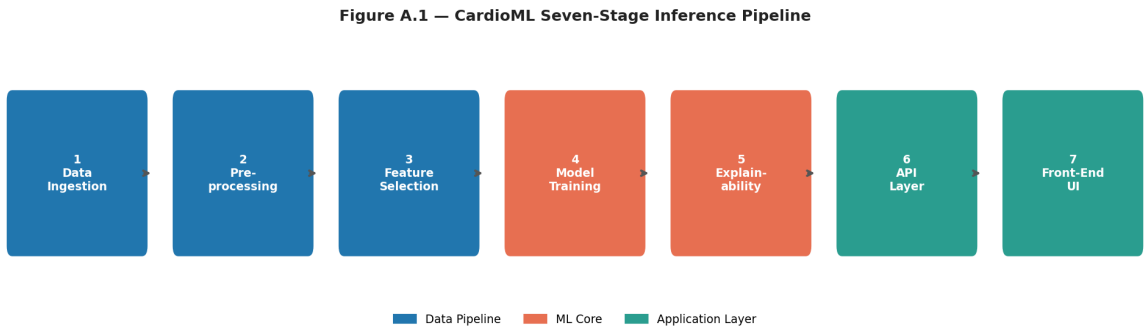

Figure A.1: CardioML seven-stage inference pipeline. Each stage is independently testable; the LR Model.pkl and Scaler.pkl artefacts are loaded once at FastAPI startup and reused across all prediction requests.

The deployed system implements a seven-stage request-response pipeline, as illustrated in Figure A.1:

1. The user completes the eight-field React input form using plain-English labels. Input values are validated client-side against permissible ranges before submission.
2. The validated form data is submitted as an HTTP POST request to the FastAPI `/predict` endpoint.
3. Pydantic schema validation verifies all eight feature values against defined data types and permissible clinical ranges, returning HTTP 422 Unprocessable Entity for any out-of-range input.
4. The valid input vector is transformed by the loaded `scaler.pkl` StandardScaler, applying the training-set-fitted scaling parameters.
5. The scaled vector is passed to `heart_risk_model.pkl`, the serialised Logistic Regression classifier, which returns the predicted risk class (Low = 0, Medium = 1, High = 2) and the corresponding class probability vector.
6. SHAP attribution is computed for the prediction using `shap.LinearExplainer`, and the ranked attributions are translated into natural language by the `explain.py` module.
7. The risk class, class probabilities, natural language explanation, and SHAP attribution vector are returned as a JSON response and asynchronously persisted to

MongoDB via the Motor driver, without storing any personally identifiable information.

The complete request-response cycle satisfies the sub-500ms response time requirement (NFR1) under standard single-user deployment conditions, and is designed for async concurrency under moderate concurrent load through FastAPI's async framework and Motor's non-blocking database operations.